Local Control over Trade: Evidence from Water Market Integration

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Abstract

Market integration directs resources toward higher-value uses, but the resulting spatial reallocation of economic activity can generate local spillovers. Communities may anticipate these spillovers and exert protectionist local control over trade. We study the integration of the world's largest water market to measure local spillovers, estimate communities' value for local production, and quantify how community control would limit trade. We find that a 10% decrease in local irrigation reduces household incomes by 3%, agricultural employment by 7%, and population by 5%. We use a dynamic residential choice model to estimate that median community willingness to pay for the local benefits of irrigation is AUD $119 per megaliter (ML), with 16% of community valuations above the average market price of $235/ML. We combine these estimates with farmer gains from trade to show how information, targeting, and the surplus available for compensation shape the effect of community control on trade. When community values are privately known and transacting farmers must secure community consent through compensation, only 62% of trades that increase farmer surplus proceed. Uniform compensation fails to target heterogeneous values and further limits trade.

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1 Introduction

Water markets offer a powerful tool for climate adaptation by reallocating scarce supplies toward higher-value uses (Rafey, 2023, Bruno & Jessoe, 2024). Yet, globally less than 1% of surface water entitlements are traded. Resistance to water leaving a community is a pervasive constraint on trade (Hanak, 2005, Grafton et al., 2011). Local governments and irrigation districts often have the power to restrict water exports or condition them on their effects on the local economy.[1] These protections reflect a central conflict: water trade creates gains by reallocating a productive resource, but that same reallocation can move economic activity away from communities that historically depended on it. When gains from market integration leave some communities behind, the resulting political backlash can threaten integration itself (Autor et al., 2020, Colantone & Stanig, 2018).

This conflict extends far beyond water markets. Market integration creates gains by reallocating economic activity toward higher-value uses, but the resulting losses can be concentrated in particular places. Import competition can shift production away from exposed communities, with persistent effects on local employment and earnings (Autor et al., 2013). Plant relocation and changes in natural-resource production similarly generate substantial local effects on employment, wages, and economic activity (Montag, 2026, Allcott & Keniston, 2018). To sustain integration in these environments, market institutions must preserve gains from trade when affected communities may develop the power to block reallocation.

In this paper, we study Australia's integration of the world's largest water market to measure local spillovers from spatial reallocation of production, estimate communities' value for local production, and quantify how community control would limit market performance. In our framework, markets direct production toward locations where it generates greater private returns, while communities share in the benefits of production occurring locally. Because these local benefits are not represented in the transactions that reallocate production, communities may seek control over trade to protect them. How much this control limits trade depends on the magnitude and heterogeneity of community values relative to private gains from trade, and on their spatial relationship. We use a dynamic residential choice model that leverages changes in migration after market integration to recover the magnitude and heterogeneity of community values, which we compare with private gains from reallocating water. We then combine these values to evaluate alternative forms of community control and compensation and analyze how their performance depends on the information available to target compensation and the surplus available to finance it.

Australia's water-market reform provides an unusually rich setting to study these forces. In 2007, sweeping reforms integrated water markets across the Murray–Darling Basin, leading to a

[1] California law, for example, conditions access to public conveyance capacity on transfers not "unreasonably affecting the overall economy" of the source county (California Water Code §1810). Some counties have adopted additional restrictions on water exports to protect agricultural economies (Hanak, 2005). Local institutional restrictions are also widespread internationally: before Australia's water-market reforms, irrigation districts could restrict water trades (Australian Competition and Consumer Commission, 2009); similar restrictions remain common in the American West (Culp et al., 2014); and Spain subjects transfers to basin-level institutional approval (Calatrava & Martínez-Granados, 2015).

nearly twenty-fold increase in the volume of permanent water entitlements traded. Water is a measurable productive input whose movement across locations can be observed, allowing us to trace how market integration reallocates production across space. The sharp increase in trade following the reform unlocked persistent differences between historical water allocations and the locations where water generated the greatest private returns. Natural soil characteristics that affect suitability for irrigation create differential exposure to market integration that we use throughout the paper to estimate its effects on communities.

The analysis proceeds in two parts. First, we measure the local consequences of market integration and estimate communities' willingness to pay to retain local production, which we compare with the private gains from reallocation. Second, we combine these estimates to quantify how alternative forms of community control and compensation affect trade.

We first document new facts about the impact of water reallocation on local economic and social outcomes. Market integration generated substantial gains and losses in irrigation across communities: persistent post-reform changes at the 10th and 90th percentiles were approximately -37% and 26% of applied irrigation. These changes align strongly with predetermined soil characteristics, providing differential exposure to reform across communities. Using this variation, we establish that water reallocation has consequences well beyond the farmers participating directly in water markets. A 10% reduction in local irrigation reduces household income by 3.0%, housing costs by 5.1%, and agricultural employment by 7.5%. Households also migrate in response to water losses, with population falling by 4.5%. Consistent with these local consequences generating opposition to reallocation, voters shift against the party responsible for the reform. These results show that water reallocation affects communities along multiple margins that are difficult to enumerate and value individually. For local control, the key economic object is instead the overall value communities place on retaining irrigation.

We recover communities' willingness to pay for the local benefits of irrigation from households' revealed residential choices. We model these choices in a dynamic discrete-choice framework (Almagro & Dominguez-Iino, 2025, Davis et al., 2023). Households are forward-looking, face costs of moving, and become more attached to a location as they remain there. They value local income and housing costs, their own tenure in a location, and the tenure of the surrounding community. Water can affect household welfare through each of these margins as well as through other unobserved local attributes. We let the effect of irrigation on household utility vary across locations. The model therefore allows us to recover a location-specific measure of the value residents place on irrigation without separately observing and monetizing each of its consequences.

We estimate the model using origin-to-destination migration flows from 2006 to 2021, separately by households' prior tenure. These choices reveal the relative flow utility of living in each community over time while accounting for moving costs and attachment to place. We then apply the same differential exposure to reform used above to these recovered flow utilities. This identifies the average effect of irrigation on the value of living in a community. To recover heterogeneity, we exploit repeated observations of communities after the reform: communities with larger water changes

should exhibit greater dispersion in flow-utility changes when the value of water differs persistently across locations. Cross-period residual variation allows us to estimate persistent heterogeneity in community values and recover the distribution of location-specific valuations even though we cannot estimate a separate causal water effect for every community.

We find that these local valuations are economically substantial. Aggregating household valuations by location, median willingness to pay for irrigation is AUD \$119/ML, with 15.8% of locations valuing water above the average market price of \$235/ML. We estimate that local irrigation's effect on household income accounts for 24.2% of median value. Local valuations are also highly heterogeneous, ranging from approximately \$67/ML at the 25th percentile to \$191/ML at the 75th percentile. The magnitude and heterogeneity of these valuations determine the compensation communities may require to permit water transfers. Whether that compensation can be supported by gains from trade also depends on the spatial correlation between community valuations and the private value of water that drives market reallocation.

To evaluate whether community losses can be accommodated by the gains from trade, we compare community values with the private productive values that motivate water reallocation. Drawing on the agricultural production model and estimates of Rafey (2023), we construct location-specific marginal productive values of water in our geography. Productive values are highly heterogeneous, with a median of approximately \$265/ML and an interquartile range of \$137–\$840/ML.

Community values are substantial relative to these productive values and similarly heterogeneous, but the two are nearly spatially uncorrelated. Their correlation is just 0.04. Consistent with the allocative objective of the reform, water moves toward locations with higher productive values, while community values do not predict where water moves. The market therefore reveals where private gains from reallocation can be created, but provides little information about where compensation is most valuable.

We combine these values to quantify how community control affects trade. When selling communities can block transfers, a trade proceeds only if the surplus available from reallocation is sufficient to secure community consent. We vary two features that determine how restrictive this constraint is: the surplus available for compensation and the information available to target it.

We first compare compensation financed solely from gains to the transacting farmers with compensation that can also draw on benefits accruing to the buying community. We then compare full information about selling-community values with an asymmetric-information environment in which communities privately know their values. Under full information, gains to transacting farmers are sufficient to fully compensate selling communities for 73% of otherwise profitable trades. Mobilizing gains to the buying community increases this share to 90%. Private information substantially reduces the share of trade that can be preserved. When transacting farmers make optimal take-it-or-leave-it offers to communities with privately known values, only 62% of trades proceed using private gains alone and 77% proceed when buying-community gains can also contribute. Our results provide a mechanism for the limited trade observed in markets subject to community consent: combined with asymmetric information, local approval can eliminate nearly 40% of otherwise profitable

trades.

Finally, we compare our bargaining protocol with a uniform compensation market institution where a fixed uniform price is offered to all selling communities and they accept or reject based on their values. Uniform compensation fails to target heterogeneous community values and reduces feasible trades by over 10 percentage points. The extent to which community control eliminates gains from trade depends critically on the surplus available for compensation and the information available to target it.

Our results make three contributions, concerning the political economy of market integration, the design of water markets, and the measurement of heterogeneous local welfare effects.

We contribute to a large literature on the distributional effects of trade. Existing work shows that market integration can reallocate production across places, generating persistent effects on local employment, earnings, and economic activity (Autor et al., 2013, Allcott & Keniston, 2018, Montag, 2026). These concentrated local effects can in turn generate political demand for protection from further reallocation (Autor et al., 2020, Colantone & Stanig, 2018, Kuziemko et al., 2026). Existing evidence has largely come from changes in product-market competition, labor demand, and the location of production. We provide new evidence from the integration of a natural-resource market, where trade directly reallocates a productive input across communities. We measure how much affected communities value retaining local production and study how markets perform when they have control over reallocation. This shifts the focus from documenting the local incidence of market integration to asking how those local consequences feed back into the extent of integration itself. By measuring community values alongside the private gains from reallocation, we quantify when compensation can reconcile local interests with gains from trade, and when community consent instead prevents otherwise profitable reallocation.

Water markets provide a particularly useful setting in which to study this problem. Research on water markets has established substantial gains from reallocating water toward higher-value uses and documented the institutions that facilitate or impede this reallocation (Young, 1986, Hanemann, 2005, Regnacq et al., 2016, Rafey, 2023, Hagerty, 2023, Ferguson, 2025). Yet much less is known about the consequences for communities from which water leaves or the value residents place on avoiding them, despite these concerns being central to opposition to water trade.[2] We quantify these local consequences and community values and compare them with the private gains that drive water reallocation. This allows us to study a central constraint on water markets—community resistance to water leaving—within the same framework used to measure their gains from trade.

Finally, spatial equilibrium and revealed-preference methods use the location choices of households to value local amenities and changes in the characteristics of the place (Bayer et al., 2007, Diamond, 2016, Almagro & Dominguez-Iino, 2025, Kashner, 2026). We extend these methods to recover heterogeneity in the welfare effects of a causally identified local attribute without requiring separate causal variation for every location. Repeated observations of communities allow us to

[2] Ge et al. (2023) estimate local impacts in a case study of a particular water trade, while Sum (2025) analyzes hypothetical local impacts of groundwater reallocation.

distinguish persistent heterogeneity in the effect of water from transitory shocks and recover the distribution of community values. This distribution, rather than only an average willingness to pay, allows us to study how heterogeneity and information about local values shape compensation and market reallocation.

The paper proceeds as follows. Section 2 describes our analytical framework. Section 3 introduces the institutional setting and data. Section 4 documents new facts about the local consequences of water market reform. Section 5 introduces our model and Section 6 describes how it is estimated. Section 7 presents parameter estimates and compares community and farmer values for irrigation. Section 8 presents counterfactuals about the impact of local control on trade. Section 9 concludes.

2 Framework

Market integration allows property-right holders to realize gains from trade by reallocating assets and resources toward higher-value users. However, the effects of trade can extend beyond the transacting parties. The transfer of productive assets may shift economic activity across communities, development can affect nearby residents, and changes in land use can alter local amenities. When these effects fall on parties outside the transaction, who holds the right to allow trade can shape the extent of market reallocation.

Realizable Trades. Consider a seller who holds a transferable good and a buyer who would like to acquire it. Let V_s^P and V_b^P denote their respective private values. The private gains from trade are $G_{sb}^P = V_b^P - V_s^P$, so reallocation creates private surplus whenever $G_{sb}^P > 0$. In our application, the good is a property right to water and these private values are the agricultural profits of the selling and buying farmers.

The transaction may also affect people who do not participate in it, who we refer to as the *community*. When the transaction shifts economic activity across places, let V_s^C denote the selling community's aggregate willingness to pay to retain the asset locally and V_b^C the buying community's aggregate willingness to pay for the local benefits of receiving it.[3]

In the free market benchmark, the buyer and seller hold the rights relevant to the transaction. They can trade whenever $G_{sb}^P > 0$ without obtaining the consent of affected communities. If instead, the selling community can block a transfer, the buyer and seller can use their gains from trade to secure their consent through compensation (Coase, 1960). If the selling community's value is known, the traders can fully compensate them from their private gains whenever $G_{sb}^P \geq V_s^C$.

A reallocation that imposes losses on one community may create gains for another. If the transfer benefits residents of the buying community, their value V_b^C provides an additional source of surplus that may be used to purchase the selling community's consent.[4] If this surplus can be accessed,

[3]When a transaction changes ownership or use without reallocating the asset across communities, there is instead a single affected community with willingness to pay, V_s^C, for the status quo use.

[4]Governments frequently mobilize anticipated local gains to influence the location of economic activity. State and local governments, for example, offer firms tax incentives and direct subsidies in exchange for investments and

trades may still proceed whenever $G^P_{sb} + V^C_b \geq V^C_s$.

Discussion. Heterogeneity in community values makes their spatial relationship with private gains from reallocation central to the effects of local control. If water leaving high-value communities generates large private gains elsewhere, those gains may be sufficient to compensate the communities and preserve trade. If high community values instead coincide with small gains from reallocation, community control will block more otherwise profitable trades.

The consequences of community veto rights also depend on information about community values. Under full information, communities can be paid exactly their value. When this value, V^C_s, is private information to the community, traders cannot distinguish communities that require high compensation from those willing to accept less. Higher offers make consent more likely but leave less of the gains from trade to the transacting parties. This information problem allows some communities to capture rents and can prevent trades that could fully compensate the community under full information (Myerson & Satterthwaite, 1983).

Alternatively, a market designer could require compensation without giving affected communities the right to block transactions. Governments could consider a uniform compensation policy under which each selling community receives a posted price when a transfer occurs. Such a policy does not require eliciting each community's value as a condition for trade, but it does not ensure that communities are fully compensated. A payment sufficient to compensate communities with high V^C_s overcompensates those with lower values, while a lower payment leaves some communities undercompensated.

The market consequences of community control depend on the magnitude and heterogeneity of community values, their relationship with the gains from reallocation, the information available to target compensation, and the sources of surplus that can be mobilized to secure consent. The empirical analysis that follows measures these values and their spatial relationship; Section 8 then uses them to quantify how alternative forms of community control affect market reallocation.

3 Institutional Background and Data

3.1 Water in the Murray–Darling Basin

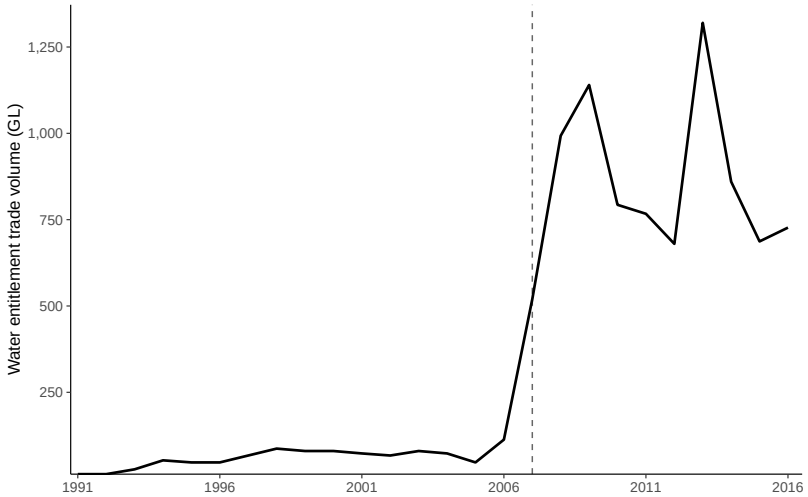
The Murray–Darling Basin (MDB) is Australia's largest and most economically significant river system, supporting around 2 million residents. It is the same size as Texas and California combined, and roughly twice the size of France. In Appendix Figure A2, we present a map of the MDB which spans parts of Queensland, New South Wales, Victoria, the Australian Capital Territory, and South Australia.

This rural agricultural region is responsible for 40% of Australia's production in irrigated agriculture. The critical agricultural input is surface water, which is highly variable with annual inflows

job creation (Slattery, 2025); more broadly, place-based policies use public resources to attract or retain economic activity because of its anticipated benefits to local residents (Kline & Moretti, 2014).

fluctuating by more than a factor of ten across years (Grafton & Wheeler, 2018). The region's aridity, volatility, and agricultural dependence have made water management a central and contested policy issue. For much of Australia's modern history, water access was managed by a fragmented system of licenses and water rights that allowed farmers to extract water from local rivers, canals, and irrigation districts. These rights were often tied to land and historical use. In the late 20th century,[5] while infrastructural and legal advances allowed for larger-scale water management, water rights remained highly heterogeneous, incompatible across state boundaries, and rarely traded in formal markets.

Figure 1: Water entitlement trade volumes over sample period



Notes: Figure reports the annual volume of water entitlement trades in the MDB. Entitlements are permanent rights to a share of available water. Data are from ABARES (Goesch et al., 2019). The dashed line denotes 2007, the beginning of the post-reform period. One GL equals one billion litres, or 1,000 megalitres (approximately 811 acre-feet). For scale, approximately 10,000-12,000 GL are delivered in an average year in the MDB.

These limitations became increasingly salient during the Millennium Drought (1997–2009), one of the worst droughts in Australian history. As inflows reached record lows and irrigation allocations fell to near zero in many regions, political support grew for Basin-wide reform. Beginning with the Water Act of 2007, a series of reforms created the most active water market in history by standardizing water rights and trading rules and integrating the market across the Basin. An important component of this integration was limiting local institutions' ability to block water from leaving their jurisdictions. New water-market rules restricted irrigation districts' ability to impede out-of-district transfers and capped the exit fees they could charge departing entitlement holders. Market integration therefore involved not only making water rights more tradable, but also curtailing local control over whether water could leave a community. Figure 1 shows the resulting rise in water-right transfers, with post-reform trade volumes reaching nearly twenty times their pre-reform levels. The reforms also facilitated large environmental water buybacks, contributing to a substantial spatial

[5] In 1994, the Council of Australian Governments (COAG) adopted a national water reform framework that promoted the separation of water rights from land, which was a key foundation for future water marketing.

reallocation of irrigation activity across the Basin.

Proponents of the reforms emphasized the efficiency and sustainability benefits of water trade. Rafey (2023) finds that post-reform allocations shifted water toward more productive agricultural users, creating value equivalent to 8–12% of total annual supply. The movement of water out of some regions, however, generated concerns about the consequences for selling communities. These concerns fueled protests and political opposition to the reforms and generated calls to restore restrictions on trade or compensate affected communities. The Australian experience reaffirmed skepticism in other parts of the world about market reallocation of water and illustrates the central tension of this paper: removing local barriers can facilitate productive reallocation while creating local consequences that generate demand for greater control over that reallocation.

3.2 Data

We combine satellite measures of water use with administrative and Census data on households, local economic activity, migration, and agricultural production. Our primary unit of analysis is the Statistical Area Level 1 (SA1), the smallest standard geography for which the Australian Census releases detailed population data. SA1s generally contain between 200 and 800 residents and average roughly 400 residents.[6] Our Murray–Darling Basin sample contains 6,494 SA1s nested within 383 Statistical Area Level 2 (SA2) regions, which we use when finer information is unavailable.

3.2.1 Water use and agricultural conditions

Evapotranspiration. Our primary measure of local water use is actual evapotranspiration (ET) from the Terrestrial Ecosystem Research Network's CMRSET product from 2000–2020. The data provide monthly estimates at 30-meter resolution across Australia. CMRSET combines remotely sensed vegetation information from high-resolution satellites with higher-frequency satellite observations and gridded meteorological data to produce continuous monthly estimates of actual evapotranspiration. We average pixel-level estimates within each SA1 and construct alternative measures that exclude pixels within 100, 250, and 500 meters of major surface waterways. Our preferred specification excludes pixels within 500 meters of waterways.

Climate. We supplement ET with gridded rainfall and potential evapotranspiration from the Australian Bureau of Meteorology from 2000–2020. These variables capture local meteorological conditions that generate evapotranspiration in the absence of irrigation.

Administrative water data. We collect annual administrative data on water allocations and trade from the Murray–Darling Basin Authority for 2000–2020. These data are reported at water-management geographies substantially larger than an SA1. We use them to validate our satellite-based measure of irrigation activity.

[6] SA1s are roughly comparable in scale to U.S. Census block groups.

Soil. We obtain predetermined soil characteristics from the Soil and Landscape Grid of Australia. The Grid combines soil observations, laboratory measurements, remote sensing, and digital-soil-mapping methods to predict non-anthropogenic soil attributes on an approximately 90-meter grid. We use measures of soil depth and available water capacity to construct instruments for post-reform changes in irrigation.

3.2.2 Households and local economic outcomes

Census outcomes. The Australian Census provides SA1-level measures of population, household income, rents, mortgage payments, employment, and residential tenure every five years from 2001 through 2021. We supplement these data with archived 1991 and 1996 Census geography files obtained through the Australian Urban Research Infrastructure Network (AURIN) when available. Appendix Table A3 reports summary statistics for key Census variables.

Migration and tenure. For the 2006, 2011, 2016, and 2021 Censuses, the Australian Bureau of Statistics (ABS) reports individuals' current residence together with their residence one and five years earlier. We observe these migration histories by age group and use them to construct annual migration choices, household tenure states, and the local stock of long-term residents used in the residential-choice model.

Agricultural production. The ABS reports agricultural production at the SA2 level at approximately five-year intervals between 2005 and 2020. For each crop, we observe the value of agricultural output, cultivated land, and applied irrigation water. We combine these data with agricultural employment and wages to adapt the production model of Rafey (2023) to our geographic units and recover location-specific marginal productive values of water. For estimation, we aggregate crops into three categories: annual crops, perennial crops, and dairy/pasture.

Additional local outcomes. We supplement the Census with SA2-level data on business activity, mortality, suicide, and federal-election outcomes. These outcomes allow us to examine whether the local effects of water reallocation extend beyond population, income, and housing markets.

4 Local Consequences of Water Reallocation

We first examine how market integration changed the spatial allocation of irrigation and the consequences of this reallocation for affected communities. We show that the reforms generated substantial and persistent shifts in irrigation activity across the Basin. Predetermined differences in agricultural suitability strongly predict the direction of these shifts, providing differential exposure to reform-induced water reallocation. We use this variation to estimate the effects of water gains and losses on local economic and social outcomes. Later in the paper, we use the same source of variation to estimate households' overall valuation of local irrigation.

4.1 Water Reallocation Following Market Integration

Measuring local irrigation. While Figure 1 shows the sharp increase in water trading after the reform, it does not reveal the resulting spatial reallocation of water. Because granular administrative data on irrigation are unavailable, we construct a local measure of irrigation activity using satellite evapotranspiration (ET) and weather data. Our preferred measure residualizes monthly ET per hectare against current and lagged rainfall, potential evapotranspiration, and SA1-by-calendar-month fixed effects, excluding pixels within 500 meters of major waterways. We define annual local irrigation, W_{jt}, as the sum of residual ET over the October–March dry season, expressed in ML per hectare. We validate this measure against administrative water data: an additional ML of water allocation is associated with approximately 0.65 ML of dry-season residual ET, while allocations have essentially no relationship with wet-season residual ET.[7] Appendix Section A describes the construction and validation exercises in detail.

We measure a community's exposure to water reallocation as the change in average irrigation activity since the reform relative to its pre-reform average,

$$\Delta W_{j,t} = \overline{W}_{j,\text{post}\leq t} - \overline{W}_{j,\text{pre}}, \tag{1}$$

allowing exposure to evolve as water is progressively reallocated across the Basin. Panel (a) of Figure 2 maps this measure for $t = 2021$, which corresponds to the average post-reform change in irrigation activity relative to the pre-reform period. The median community experienced an absolute change of 0.15 ML/ha, equivalent to approximately 16% of applied irrigation; at the 10th and 90th percentiles, changes were approximately −37% and 26%, respectively. The median change was modestly negative, consistent with the substantial volume of water purchased through the government's environmental water buyback program.

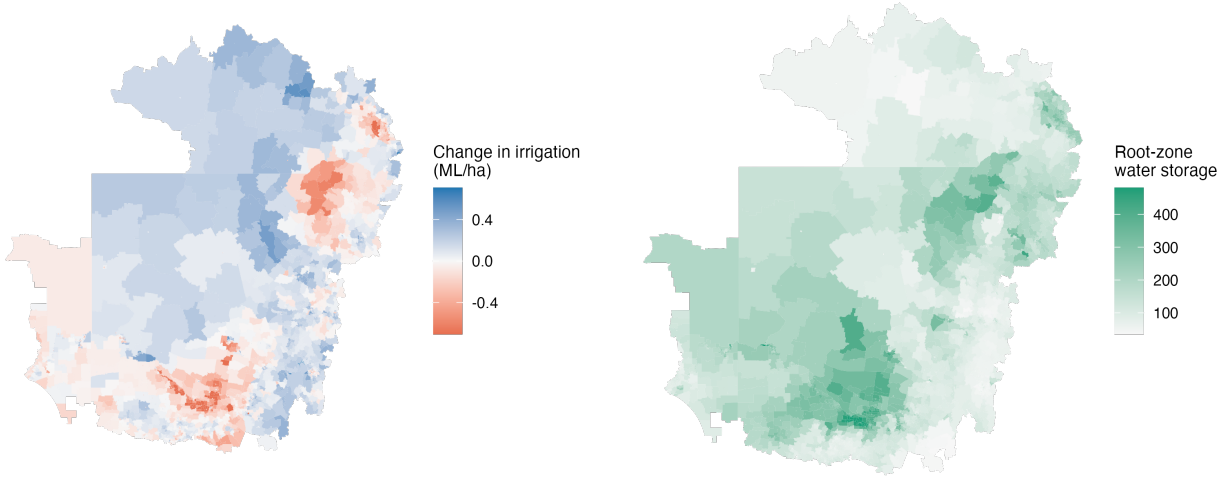
Differential exposure to reform. The substantial gains and losses in irrigation documented above provide differential exposure to market integration across communities. To predict this reallocation, we use root-zone water storage, a predetermined soil characteristic constructed as the interaction of available water capacity and soil depth. Root-zone water storage captures the amount of water that can be stored within the soil profile and made available to plant roots. Soils with lower root-zone storage may still be productive, but depend more heavily on reliable irrigation supply. Panel (b) of Figure 2 maps root-zone water storage across the Basin. Comparing Panels (a) and (b) reveals a clear spatial relationship: areas with lower root-zone water storage tend to experience larger increases in irrigation following market integration.

We quantify this relationship by estimating

$$\Delta W_{j,t} = \pi_t Z_j + \gamma_{d(j)} + \nu_{j,t}, \tag{2}$$

[7] This magnitude is consistent with estimates of irrigation efficiency in the MDB: Roth et al. (2013) report whole-farm irrigation efficiencies of approximately 57–70%.

Figure 2: Water Reallocation and Predetermined Agricultural Suitability



(a) Change in irrigation activity

(b) Root-zone water storage

Notes: Panel (a) maps the change in local irrigation activity between the pre-reform period and 2021, measured in ML per hectare. Panel (b) maps root-zone water storage, constructed as the interaction of available water capacity and soil depth. Color scales are capped at the 1st and 99th percentiles to limit the influence of extreme values.

where Z_j is root-zone water storage and $\gamma_{d(j)}$ denotes water-district fixed effects. Root-zone water storage negatively predicts changes in irrigation in each post-reform period, consistent with irrigation expanding relatively more on soils that depend more heavily on reliable water supply. The relationship is strong, with F-statistics ranging from 28.0 to 35.6 across post-reform years. Appendix Table A4 reports the estimates. The interaction of this predetermined soil variation with market integration provides plausibly exogenous variation in local irrigation, which we use below to estimate the effects of water gains and losses on affected communities.

4.2 Local Economic Consequences

We next examine how the reallocation of water across the Basin affects the communities that gain and lose irrigation. We use the differential exposure documented above to estimate these local effects. For outcome Y_{jt}, we estimate

$$\Delta Y_{j,t} = \beta_t \Delta W_{j,t} + \gamma_{d(j)} + \varepsilon_{j,t}, \tag{3}$$

where $\Delta Y_{j,t}$ is the change in the outcome relative to its pre-reform baseline and $\Delta W_{j,t}$ is the corresponding change in local irrigation. We instrument $\Delta W_{j,t}$ using the predetermined soil variation described in Equation (2). In the post-reform period, water exposure is measured through the cor-

responding outcome year, allowing the treatment to evolve as irrigation is progressively reallocated across the Basin. Standard errors are clustered at the SA2 level.

For outcomes observed before market integration, we use the same design to test whether communities that would subsequently experience different water changes were already on different trajectories (Alsan et al., 2025). In these pre-reform specifications, we use eventual post-reform water reallocation as the treatment and instrument it with the same predetermined soil variation. Identification requires that this variation affect changes in local outcomes following market integration through its effect on irrigation rather than through other differential changes across communities. The pre-reform estimates then provide a direct test for differential trends associated with our instrument.

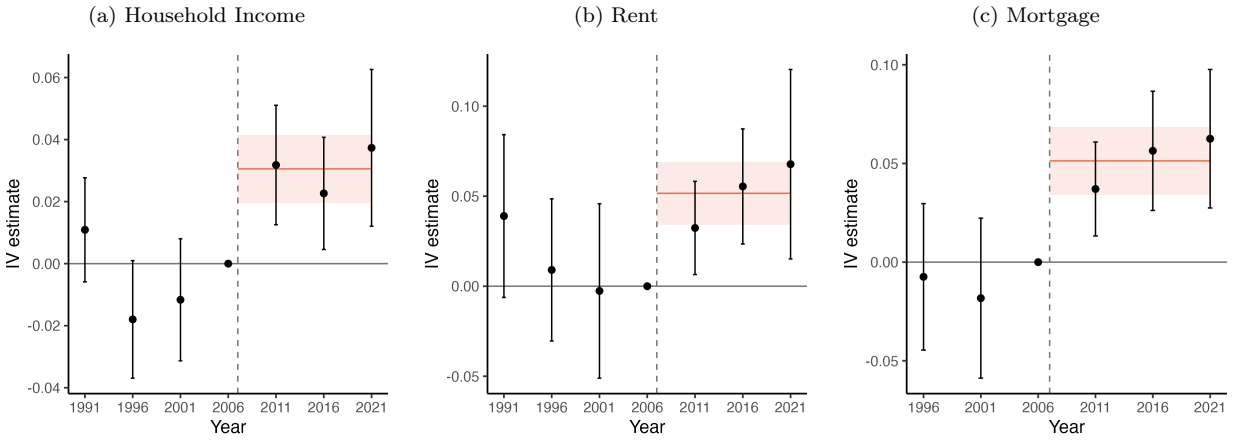
Local economic activity. Figure 3 reports effects on the three economic outcomes that we observe both before and after the reform: household income, rent, and mortgage payments. We find immediate and persistent effects on all three. A 10% reduction in local irrigation reduces household income by approximately 3.0% and housing costs by approximately 5.1%, with similar estimates across post-reform years. The pooled estimates summarize these persistent effects. The longer panels for these outcomes also allow us to evaluate pre-trends. We find no evidence that communities subsequently exposed to water gains or losses were already experiencing differential trends in income or housing costs before the reform.

For outcomes available only after the reform, we estimate pooled specifications using the same instrument-induced variation in irrigation. Appendix Table A5 reports the pooled estimates for all outcomes discussed in this subsection. A 10% reduction in irrigation reduces agricultural employment by approximately 7.5% and the number of operating businesses by 2.6%. We estimate a precise near-zero effect on personal incomes, so the reform-induced change in irrigation seems to affect household incomes through employment, rather than through wages.

Household migration and social response. Households respond to these changes in local economic conditions. A 10% reduction in irrigation reduces the population by 2.7% in 2011 and 7.1% by 2021, with a pooled effect of 4.5%. We find a similarly gradual response in the stock of long-term residents, a measure of the persistence of the local community: the effect grows from 1.3% in 2011 to 6.9% by 2021, with a pooled effect of 3.7%. We depict these transitions in Appendix Figure A3. Evidence of this gradual response relative to local economic indicators motivates migration frictions and a dynamic response to reform for our residential choice model in Section 5.

Water loss also produces political and social responses. A 10% reduction in irrigation increases the Labor vote share by approximately 2.1 percentage points and suicides by 17.4%. These results reinforce that water reallocation affects communities along dimensions beyond the economic outcomes observed above.

Figure 3: Effect of Water Gain on Income, Rent, and Mortgage



Notes: Figure depicts the effect of a persistent increase of 0.096 ET/hectare on SA1 incomes, rents, and mortgage payments relative to 2006 for each Census year from 1996-2021. We instrument for water change with soil characteristics. One unit of water change is equivalent to a 10% increase in irrigation intensity at the average SA1. Annual estimates of the effect are reported in black dots and the pooled post-reform effect is the horizontal red line. Standard errors are clustered at the SA2 level for annual results and at the SA1 level for the pooled effect.

4.3 From Local Consequences to Community Values

The results above show that water reallocation affects communities along multiple margins, from income and housing costs to migration, political behavior, and social distress. Many of these consequences cannot be directly expressed in dollars, yet the framework in Section 2 requires an encompassing measure of residents' willingness to pay to retain water locally. At the same time, the reduced-form results provide support for our empirical design: predetermined soil variation strongly predicts post-reform water reallocation, but does not predict differential trends in the outcomes we observe before the reform. We therefore use this same variation in a dynamic residential choice model, recovering community willingness to pay from households' decisions to remain in, leave, or move to communities as local irrigation changes.

5 Model

We recover the value that residents place on retaining the local benefits of irrigation from their revealed residential choices. Households reveal their preferences over communities through decisions to remain, leave, or move into them. Because these decisions are persistent and forward-looking, we model them using a dynamic residential choice framework. The model allows water to affect the attractiveness of a community through observed economic and demographic conditions, as well as unobserved local attributes. We use the resulting flow utilities to define the community value of water. We then define the productive value of water from agricultural producers' profit maximization, providing the private-value benchmark for our market-design analysis.

5.1 Residential Choice

There are $J+1$ locations: J SA1s in the Murray–Darling Basin, indexed by $j \in \mathcal{J}$, and an outside option, $j = 0$, representing the rest of Australia. At the beginning of each period, household h chooses where to live. Its individual state is

$$x_{ht} = \{j_{h,t-1}, \tau_{h,t-1}\}, \tag{4}$$

where $j_{h,t-1}$ is the household's previous location and $\tau_{h,t-1}$ is its tenure in that location.

Flow utility. The flow utility household h receives from choosing location j is

$$u_{ht}(j) = \overline{u}_{jt\tau} + MC_t(j, j_{h,t-1}) + \varepsilon_{hjt}, \tag{5}$$

where τ denotes the household's tenure state, $\overline{u}_{jt\tau}$ is flow indirect utility from residing in location j, $MC_t(\cdot)$ captures moving costs, and ε_{hjt} is an independently and identically distributed Type-I extreme-value preference shock.

We parameterize flow indirect utility as

$$\overline{u}_{jt\tau} = \alpha_I \log I_{jt} + \alpha_p \log p_{jt} + \alpha_W W_{jt} + \alpha_T T_{jt} + \lambda_\tau + \xi_{jt\tau}, \tag{6}$$

where I_{jt} is household income, p_{jt} is the rental price, W_{jt} is local irrigation, and T_{jt} is the stock of long-term residents. The tenure-state fixed effects λ_τ capture differences in utility associated with a household's own tenure in a location, while α_T captures households' preference for living in communities with a larger stock of long-term residents. The term $\xi_{jt\tau}$ captures other unobserved determinants of local utility.

This specification distinguishes a household's own attachment to place from its preference for the persistence of the surrounding community. It also does not require us to enumerate every consequence of water reallocation. Water may affect utility directly, through income and the persistence of the local community, or through other valued attributes of place captured by $\xi_{jt\tau}$. The community value we recover below incorporates all of these channels.

Dynamic location choice. Residential choices exhibit substantial persistence: households move infrequently, and the probability of leaving a community varies systematically with how long a household has lived there. We capture these features through moving costs and tenure-dependent flow utility. Because today's residential choice determines both the household's future location and its tenure state, households are forward-looking.

The household's value function satisfies

$$V_t(x_{ht}, \varepsilon_{ht}) = \max_j \left\{ u_{ht}(j) + \beta \mathbb{E}\left[V_{t+1}(x_{h,t+1}, \varepsilon_{h,t+1}) \mid j, x_{ht} \right] \right\}, \tag{7}$$

where $\beta \in (0,1)$ is the discount factor. A household that remains in its current location preserves and accumulates tenure, while a household that moves begins with no tenure in its new location.

The dynamic structure is important for recovering flow utility from residential choices. An observed decision to remain in a community reflects not only the utility the household receives today, but also moving costs and the consequences of the decision for future location and tenure. Section 6.1 shows how observed migration choices and a renewal structure in the household state allow us to recover $\overline{u}_{jt\tau}$ while accounting for these forward-looking incentives.

5.1.1 Community Value of Water

We use the residential choice model to measure the value households place on the consequences of local water reallocation. Irrigation can affect household utility both through its conditional effect in Equation (6) and by changing other attributes of the community. The total effect of irrigation on household flow utility is

$$\frac{d\overline{u}_{jt}}{dW_{jt}} = \underbrace{\frac{\partial \overline{u}_{jt}}{\partial W_{jt}}}_{\text{conditional water effect}} + \underbrace{\frac{\partial \overline{u}_{jt}}{\partial \log I_{jt}}\frac{d\log I_{jt}}{dW_{jt}} + \frac{\partial \overline{u}_{jt}}{\partial \log p_{jt}}\frac{d\log p_{jt}}{dW_{jt}} + \frac{\partial \overline{u}_{jt}}{\partial T_{jt}}\frac{dT_{jt}}{dW_{jt}}}_{\text{observed equilibrium effects}} + \underbrace{\frac{d\xi_{jt}}{dW_{jt}}}_{\text{other local attributes}}. \tag{8}$$

These responses may differ across communities. To make the sources of this heterogeneity explicit, write

$$\frac{d\log I_{jt}}{dW_{jt}} = \beta_I + \eta_j^I, \qquad \frac{dT_{jt}}{dW_{jt}} = \beta_T + \eta_j^T, \qquad \frac{d\xi_{jt}}{dW_{jt}} = \eta_j^W, \tag{9}$$

where the η_j terms capture community-specific departures from the average response. We normalize each heterogeneous component to have mean zero across communities.

We assume that changes in local housing costs affect residential choices, but we treat changes in housing expenditures induced by water reallocation as transfers between residents and local property owners rather than as a net cost to the community.[8]

We define non-housing flow utility as

$$\zeta_{jt\tau} \equiv \overline{u}_{jt\tau} - \alpha_p \log p_{jt} - \lambda_\tau. \tag{10}$$

The effect of irrigation on this utility index, $\frac{d\zeta_{jt}}{dW_{jt}}$, contains each of the channels in Equations (8) other than the housing-transfer component. We denote this encompassing effect of irrigation on community utility by ω_j which we summarize with a mean and location-specific components as

$$\omega_j \equiv \frac{d\zeta_{jt}}{dW_{jt}} = \overline{\omega} + \eta_j, \qquad \mathbb{E}[\eta_j] = 0. \tag{11}$$

The component $\overline{\omega}$ is the mean utility effect of irrigation and η_j captures heterogeneity in community values. We do not estimate ω_j or η_j by separately estimating and summing these channels. Instead,

[8]This treatment assumes that changes in housing expenditures induced by water reallocation accrue to property owners within the community. We nevertheless retain housing prices in household utility because they affect residential choices and, below, provide the numeraire used to express utility values in monetary terms.

our empirical strategy estimates their total effect on local utility.[9] Section 6.1.3 shows how repeated reform-induced changes in water allow us to recover the distribution of this total community value even when location-specific causal effects cannot be estimated individually.

We sum up household ω_j to obtain aggregate community value by

$$V_{jt}^{C} = M_{jt}\omega_j, \tag{12}$$

where M_{jt} is the number of households in community j. This value is initially expressed in utility units. We use households' estimated housing-price preferences to translate it into dollars, making community values directly comparable to the private value of water.

5.2 Productive Value of Water

To compare community values with the private gains from water reallocation, we also require the marginal value of water to agricultural producers. A producer of crop c in location j produces output using irrigation water W_{jct}, rainfall R_{jct}, cultivated land K_{jct}, labor L_{jct}, materials M_{jct}, and productivity ω_{jct}. Conditional on crop choice and cultivated land, the producer chooses variable inputs to maximize operating profits:

$$\pi_{jct}(W_{jct}) = \max_{L_{jct},M_{jct}} \left\{F_{jc}(W_{jct}, R_{jct}, K_{jct}, L_{jct}, M_{jct}; \omega_{jct}) - w_{jt}L_{jct} - q_{jct}M_{jct}\right\}. \tag{13}$$

Farms are price-takers in the market for commodities, labor, and materials. By the envelope theorem, the marginal effect of water on optimized operating profits is

$$V_{jct}^{P} \equiv \frac{\partial \pi_{jct}}{\partial W_{jct}} = F_{W,jc}. \tag{14}$$

Thus, the productive value of water is its marginal contribution to farm profits after allowing producers to optimally adjust variable inputs. Section 6.1.3 describes how we estimate $F_{W,jc}$ and aggregate crop-level values to construct the location-level private values used in our comparison with community values.

6 Estimation

The framework requires two empirical objects: the value communities place on retaining water locally and the private value of water to agricultural producers. We recover community values from a dynamic residential choice model that relates households' migration decisions to the flow utility of living in each community. We construct productive values by adapting existing farm-level production estimates to our regional data.

[9] Using our causal estimates of water on local outcomes and our demand estimates, we can get a sense of which channels are contributing to the total value of irrigation in a community.

6.1 Residential Choice and Community Water Values

We use households' observed residential choices to recover the flow utility of living in each community. Our data record flows of households' previous and current locations as well as their tenure in their previous location. These migration histories allow us to distinguish the utility associated with remaining in a community from moving costs and the forward-looking value of preserving tenure.

We estimate the model using a two-step, BLP-style procedure adapted to our dynamic location-choice setting (Davis et al., 2023, Berry et al., 1995). First, we use conditional choice probabilities and the renewal structure of the dynamic problem to construct a continuation-value correction. Conditional on this correction, maximum likelihood recovers tenure-specific mean flow utilities and moving costs from observed origin-destination flows. Second, we decompose the recovered flow utilities into income, housing prices, irrigation, community tenure stock, and unobserved local amenities using instrumental variables.

6.1.1 Recovering Flow Indirect Utilities

We first derive an equation relating flow indirect utility and moving costs to observed conditional choice probabilities. Closely following Almagro & Dominguez-Iino (2025), we use a Hotz–Miller inversion together with restrictions on the evolution of household tenure. The resulting renewal structure eliminates otherwise unknown continuation values from the estimating equation. We provide the full derivation in Appendix Section B.

We begin with standard assumptions on the dynamic discrete-choice environment.

Assumption 1 (Dynamic choice environment)**.** *Households are atomistic and the household state follows a Markov process with a conditional-independence structure. In particular, an individual household's residential choice does not affect the evolution of the aggregate state,*

$$\mathbb{P}(\Omega_{t+1} \mid j, x_t, \Omega_t, \varepsilon_t) = \mathbb{P}(\Omega_{t+1} \mid \Omega_t), \tag{15}$$

while the evolution of the household-specific state depends only on the current state, aggregate state, and residential choice. Idiosyncratic preference shocks ε_{hjt} are independently and identically distributed Type-I extreme value across households, locations, and periods.

Assumption 1 gives the household problem the standard conditional-independence structure of a dynamic discrete-choice model. Current idiosyncratic preference shocks do not predict future states or preference shocks, and an individual household takes the evolution of aggregate conditions as given. Going forward, we leave dependence on the aggregate state implicit. We normalize the flow utility of the outside option to zero.

Let $\overline{V}_t(x)$ denote the expected value of entering period t in household state x, prior to the realization of idiosyncratic preference shocks. Iteratively applying the Hotz–Miller inversion, the log probability of a household with state $x_{t-1} = (j_{t-1}, \tau_{t-1})$ choosing location j relative to the

outside option j_0 can be written as

$$\begin{aligned}\log \frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} = \overline{u}_{jt\tau_t(j,\tau_{t-1})} - \overline{u}_{j_0t,0} + MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ + \beta \left[MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0)\right] \\ - \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid j, \tau_t(j, \tau_{t-1}))}{\mathbb{P}_{t+1}(j_0 \mid j_0, 0)} \\ + \beta^2 \left[\overline{V}_{t+2}(j_0, \tau_{t+1}(j_0, \tau_t(j, \tau_{t-1}))) - \overline{V}_{t+2}(j_0, 0)\right]. \end{aligned} \tag{16}$$

Equation (16) separates current flow utility and moving costs from the forward-looking consequences of the residential choice. Without further restrictions, estimation would still require solving for the continuation values in the final line. We instead exploit the structure of tenure and the composite outside option to construct a renewal path under which the two choice histories reach the same payoff-relevant state after two periods.

Renewal through the outside option. We model tenure as a payoff-relevant state only while a household resides within the Murray–Darling Basin. A household that moves to a new MDB location begins with no tenure in that location, while remaining in the same location preserves and increments tenure. Choosing the composite outside option resets modeled tenure.

Assumption 2 (Tenure and outside-state renewal)**.** *Household tenure evolves according to*

$$\tau_t(j, \tau_{t-1}) = \begin{cases} \tau_{t-1} + 1, & j = j_{t-1}, \quad j \neq j_0, \\ 0, & j \neq j_{t-1}, \end{cases} \tag{17}$$

and tenure is not modeled within the composite outside option, $\tau_t(j_0, \tau) = 0$ for all τ.

Assumption 2 makes the outside option a renewal state. Consider a household that chooses location j in period t and the outside option in period $t+1$, relative to a household choosing the outside option in both periods. Both histories arrive in period $t+2$ in the same payoff-relevant state, $(j_0, 0)$. Hence,

$$\overline{V}_{t+2}(j_0, 0) - \overline{V}_{t+2}(j_0, 0) = 0, \tag{18}$$

and all continuation values from period $t+2$ onward cancel. This finite-dependence property allows us to estimate the dynamic choice model without explicitly computing household value functions.

We parameterize contemporaneous moving costs as a linear function of distance,

$$MC_t(j, j_{t-1}) = \kappa \, \text{Dist}(j, j_{t-1}). \tag{19}$$

The renewal expression also generates moving-cost components associated with subsequently reaching the outside option. Components that vary only with the household's origin are absorbed by origin fixed effects, while destination-specific components are absorbed into the destination flow utilities recovered below. Appendix Section B provides the complete decomposition.

Observed tenure states and migration histories. For each Census year $t \in \{2006, 2011, 2016, 2021\}$, we observe a household's location five years prior, j_{t-5}, one year prior, j_{t-1}, and at the time of the Census, j_t. We treat the transition from j_{t-1} to j_t as the annual residential choice and use j_{t-5} to construct the household's tenure state.

Because locations at $t-5$ and $t-1$ are observed at the SA2 level, tenure is defined with respect to residence in the same SA2.[10] Immediately prior to the annual choice,

$$\tau_{t-1} = \begin{cases} 4+, & j_{t-5} = j_{t-1}, \\ [0,4), & j_{t-5} \neq j_{t-1}. \end{cases} \tag{20}$$

Combining this state with the observed annual choice gives

$$\tau_t(j, \tau_{t-1}) = \begin{cases} 5+, & j = j_{t-1} \text{ and } \tau_{t-1} = 4+, \\ [1,5), & j = j_{t-1} \text{ and } \tau_{t-1} = [0,4), \\ [0,1), & j \neq j_{t-1}. \end{cases} \tag{21}$$

These states capture the strong state dependence in residential mobility observed in the data: long-term residents are substantially less likely to leave their communities than recent arrivals. Together with geographic moving costs, this pattern motivates the dynamic structure of the model. An observed decision to remain cannot be interpreted solely as a preference for the community's current attributes; it also reflects the costs of moving and the continuation value associated with the household's tenure state.

These migration histories identify current annual choice probabilities conditional on tenure. The continuation correction additionally requires exit probabilities in the subsequent annual period, which we do not observe directly. We therefore impose a local stationarity approximation.

Assumption 3 (Stationary-policy approximation)**.** *The annual exit policy is locally stable between adjacent periods. For any MDB location j and tenure state τ,*

$$\mathbb{P}_{t+1}(j_0 \mid j, \tau) \approx \mathbb{P}_t(j_0 \mid j, \tau). \tag{22}$$

Let $\widehat{q}_t(j_0 \mid x)$ denote the estimated annual exit probability. We summarize the forward-looking component of the location-choice problem with the continuation correction

$$\mathcal{C}_t(j, x_{t-1}; \widehat{q}) \equiv \beta \log \frac{\widehat{q}_t(j_0 \mid j, \tau_t(j, \tau_{t-1}))}{\widehat{q}_t(j_0 \mid j_0, 0)}. \tag{23}$$

The assumptions above reduce the dynamic location-choice problem to a standard conditional-choice index with an observed continuation correction. For each inside location $j \neq j_0$, define

$$G_{jt}(x_{t-1}; \theta, \widehat{q}) = \delta_{jt\tau} + \gamma_{j_{t-1}} + \kappa \, \mathrm{Dist}(j, j_{t-1}) - \mathcal{C}_t(j, x_{t-1}; \widehat{q}), \tag{24}$$

[10] This allows households to maintain tenure within regions with an average population of approximately 10,000 even if they change residences within the region.

where $\delta_{jt\tau}$ is the flow indirect utility associated with destination j, period t, and tenure state τ, $\gamma_{j_{t-1}}$ captures origin-specific components of moving costs, and κ governs the effect of distance on moving costs.

Conditional on the estimated continuation correction, choice probabilities take the familiar logit form,

$$P_t(j \mid x_{t-1}; \theta, \widehat{q}) = \frac{\exp\{G_{jt}(x_{t-1}; \theta, \widehat{q})\}}{1 + \sum_{\ell \neq j_0} \exp\{G_{\ell t}(x_{t-1}; \theta, \widehat{q})\}}, \qquad j \neq j_0. \tag{25}$$

Let $N_t(j; x)$ denote the number of households observed in state x that choose location j. We estimate the first-stage parameters by maximum likelihood,

$$\widehat{\theta} = \arg\max_{\theta} \sum_{t,x} \sum_{j \in \mathcal{J}} N_t(j; x) \log P_t(j \mid x; \theta, \widehat{q}). \tag{26}$$

The first step uses observed migration choices to recover tenure-specific flow indirect utilities $\delta_{jt\tau}$ while accounting for moving costs and the forward-looking consequences of current residential choices. Importantly, we do not need to solve the household's value function or specify a parametric law of motion for the aggregate state. The dynamic component is summarized by the continuation correction in Equation (23).

The MLE formulation also substantially reduces the need to smooth empirical conditional choice probabilities. Many destination choices are never observed from a particular location-tenure state, but zero choice counts simply contribute zero to the likelihood. Regularization is required only for the exit probabilities entering Equation (23), since these appear inside a logarithm and must be strictly positive. We regularize these probabilities with a Beta-Binomial estimator that adds a small number of prior pseudo-counts to observed outside and inside choices. Appendix Section B.4 provides details.

6.1.2 Location Preferences

In the second step, we decompose the recovered flow indirect utilities into the location characteristics introduced in Section 5. Specifically,

$$\widehat{\delta}_{jt\tau} = \alpha_I \log I_{jt} + \alpha_p \log p_{jt} + \alpha_W W_{jt} + \alpha_T T_{jt} + \lambda_\tau + \gamma_j + \gamma_{r(j),t} + \xi_{jt\tau}, \tag{27}$$

where λ_τ are tenure-state fixed effects, γ_j are location fixed effects, and $\gamma_{r(j),t}$ are water-district-by-year fixed effects. Because income, housing prices, water use, and tenure stock may respond to unobserved changes in local amenities, we instrument for each. The water instrument is discussed in Section 4 and we discuss the construction of the tenure stock, income, and price instruments in Appendix Section C.

Water. We instrument for irrigation W_{jt} using the same predetermined soil variation interacted with water-market reform timing used in Section 4. The instrument therefore isolates reform-induced

changes in local irrigation rather than variation arising from contemporaneous local conditions.

Tenure stock. Unobserved amenities may both increase residential persistence and make a location more attractive. We instrument for T_{jt} using a demographic shift-share instrument that interacts each SA1's lagged age-by-tenure composition with age-specific residential retention rates calculated outside the SA1's own SA2. The instrument predicts changes in tenure stock from predetermined demographic composition and the propensity of similar households elsewhere to remain in place.

Income. We instrument for household income using a Bartik-style shift-share instrument that interacts baseline industry employment shares with national industry-level employment growth. Identification comes from differential exposure to national employment changes induced by pre-existing local industry composition.

Housing prices. We instrument for housing prices using predicted local mortality shocks, following Kashner (2026). We interact lagged SA1 age shares with national age-specific mortality rates, generating variation in local housing demand from predetermined demographic composition rather than contemporaneous local conditions.

Given instruments

$$Z_{jt} = [z^p_{jt}, z^W_{jt}, z^I_{jt}, z^T_{jt}],$$

and the moment condition $\mathbb{E}\left[Z_{jt}\xi^k_{jt\tau}\right] = 0$, we estimate the relevant coefficients via IV-GMM. With multiple endogenous community characteristics, identification additionally requires that the instruments generate sufficiently independent variation to separately identify their effects. Formally, the matrix of first-stage relationships between the instruments and endogenous characteristics must have full column rank.[11]

6.1.3 Community Value of Water

The model defines community water value as the encompassing effect of irrigation on local amenities. Rather than constructing this value from separate estimates of the channels in Equation (8), we estimate it directly with the same differential exposure to reform used in Section 4.

We construct the empirical analogue of the local amenity index, $\widehat{\zeta}_{jt\tau}$ defined in Equation (10) and take differences relative to the pre-reform period to get $\Delta\widehat{\zeta}_{jt\tau}$. Define $\Delta\widehat{\zeta}_{jt}$ as the tenure-state population-weighted change in household flow utility net of housing price, or the amenity value of living in j. Our definition of ω_j from Equation (11) implies

$$\Delta\widehat{\zeta}_{jt} = (\overline{\omega} + \eta_j)\Delta W_{jt} + e_{jt}. \tag{28}$$

[11] Given that there is sampling uncertainty in the first-stage MLE estimates of $\widehat{\delta}_{jt\tau}$, we must address this in our statistical inference procedures (Murphy & Topel, 2002). In Appendix Section H, we develop a bootstrap that re-estimates both stages to incorporate this source of uncertainty.

where $\overline{\omega}$ is the mean willingness to pay and η_j is the mean zero location-specific deviation from the mean. Nuisance parameters e_{jt} capture changes in local utility unrelated to the persistent community-specific value for local irrigation.

We estimate the distribution of ω_j across communities and use each community's observed utility responses to form a location-specific prediction.

Mean community value. We first estimate the mean effect of irrigated water on amenities, $\overline{\omega}$, exploiting the same variation used to estimate the local consequences of water reallocation in Section 4, now using $\Delta\widehat{\zeta}_{jt}$ as the outcome. For each post-reform wave, we regress cumulative changes in irrigation on the predetermined soil instrument, Z_j, and water-district fixed effects:

$$\Delta W_{jt} = \pi_t Z_j + \kappa_{d(j),t} + v_{jt}. \tag{29}$$

We retain the estimated residuals, $\widehat{v}_{jt}$, which measure the component of water changes not predicted by the soil instrument and district effects.

We then estimate a stacked control-function regression, allowing the coefficient on the first-stage residual to vary across waves:

$$\Delta\widehat{\zeta}_{jt} = \overline{\omega}\Delta W_{jt} + \rho_t \widehat{v}_{jt} + \gamma_{d(j),t} + r_{jt}. \tag{30}$$

The coefficient on water changes is numerically equivalent to the corresponding 2SLS estimate. We use the control-function representation because it also estimates the association between residual water changes and changes in location value, which we remove before estimating persistent heterogeneity.

Interpreting the water coefficient as the mean community value requires that the soil instrument predict irrigation changes and, conditional on district–year effects, be unrelated to non-water changes in location value. With heterogeneous effects, we additionally require that instrument-induced exposure does not systematically weight communities with higher or lower water values.Under these conditions, the coefficient identifies $\overline{\omega}$, the average effect of irrigation on location value. This encompasses irrigation's effects through income and other local attributes, without requiring us to separately estimate or monetize each channel.

Heterogeneity in community value. Household willingness to pay for water also includes the location-specific component, η_j. Individual location-specific effects cannot be estimated precisely from the small number of post-reform observations available for each community. Instead, we first estimate the distribution of persistent water effects across communities and then recover posterior estimates of each location's specific value.

Our estimator uses comparisons in residual changes in location value, r_{jt}, for 2011, 2016, and 2021 from Equation (30).[12] These residuals, after removing the estimated common water effect, control function, and district–year effects from changes in location values contain any potential

[12]Since r_{jt} represents changes in residual location value relative to 2006, $r_{j,2006} = 0$.

heterogeneous impact along with changes in other shocks to location value. In Equation (31), we specify the two components of r_{jt} by

$$r_{jt} = \eta_j \Delta W_{jt} + e_{jt} - e_{j0}, \tag{31}$$

where η_j is the persistent deviation from mean community water value and $e_{jt} - e_{j0}$ captures remaining utility changes.

To separate the heterogeneous values for water from residual utility drift, we leverage differences in amenity values across periods. For periods $t, s \in \{2006, 2011, 2016, 2021\}$ we define the differences $q_{j,ts}$ by

$$q_{j,ts} \equiv r_{jt} - r_{js} = \eta_j(\Delta W_{jt} - \Delta W_{js}) + (e_{jt} - e_{js}) \tag{32}$$

For pre-post differences, Equation (32) simplifies to $q_{j,t,2006} = r_{jt}$. Since water exposure is nearly constant between post-reform periods $(t, s) \in \{(2016, 2011), (2021, 2011), (2021, 2016)\}$, the persistent water component cancels, $\Delta W_{jt} - \Delta W_{js} \approx 0$, and post-post differences isolate the noisy non-water changes in location utility, $q_{j,ts} \approx e_{jt} - e_{js}$.

This motivates estimating the distribution of water values by deconvolving the post–post noise proxy from the pre–post signals. The distribution of post–post changes in amenities is informative about the distribution of unobservable amenity drift. The distribution of pre–post changes in amenities is informative about the combination of unobservable amenity drift *and* the distribution of heterogeneity in η_j.[13]

In Appendix Section D we provide a proof of nonparametric identification of the full distribution of η_j. This proof requires conditional independence and stability of the relevant noise distributions, together with support and invertibility conditions. The conditional independence assumption requires for our instrument Z_j to be uncorrelated with η_j and other non-water forms of utility drift.[14] We also require stability of the distribution of the evolution of the unobservable shocks, $e_{jt} - e_{js}$.

We estimate the noise distribution of non-water utility changes $u_{j,ts} = e_{jt} - e_{j0}$ nonparametrically using Gaussian kernel densities of the post–post comparisons. We estimate these densities separately across seven quantile bins of average first-stage residuals, V_j from Equation (29):

$$\widehat{\overline{v}}_j = \frac{1}{3}\left(\widehat{v}_{j,2011} + \widehat{v}_{j,2016} + \widehat{v}_{j,2021}\right). \tag{33}$$

The control function removes a linear drift component, while binning allows the remaining noise distribution to differ with residual water acquisition, or loss. Let $b(j)$ denote community j's residual bin and let $\widehat{f}_{b(j)}$ denote its fitted noise density.

[13] In Appendix Figure A4, we present the density of $q_{j,t,s}$. We observe much larger dispersion in the pre–post comparisons than the post–post comparisons, consistent with additional variance from heterogeneous η_j.

[14] The proof does not require that η_j is uncorrelated with V_j. This method can accommodate Roy-style selection on community gains. Doing so requires using V_j in a control function style approach to recover heterogeneous η_j distributions by V_j bins. In our empirical setting, we expect farmers to make individually rational trading decisions, considering their own profits and thus do not expect Roy-style selection on η_j. In Appendix Figure A1 we provide support for this, as shown by a control function approach. We find that across the support of the realizations of V_j, the estimated mean effect of changes in water on amenities is approximately constant. For this reason, we pool across bins of V_j to estimate a single population distribution.

Given these noise distributions, we estimate the distribution of location-specific deviations from mean willingness to pay for irrigation, F_η. For a candidate distribution F, the implied marginal density of a pre–post signal is

$$g_{jt}(\widetilde{r}; F) = \int \widehat{f}_{b(j)} \left(\widetilde{r} - \theta \Delta W_{jt}\right) \, dF(\theta). \tag{34}$$

This convolution averages over candidate water values, evaluating the noise realization required for each value to explain the observed response. We estimate the distribution by maximizing the marginal composite log likelihood:

$$\widehat{F}_\eta = \arg\max_F \sum_j \sum_{t \in \{2011,2016,2021\}} \log g_{jt}(\widetilde{r}_{jt}; F). \tag{35}$$

The criterion combines the marginal signals without specifying their joint distribution within a community. Our main specification estimates two-Gaussian mixing family.[15][16]

Location-specific values. We combine the fitted distribution with each community's three utility responses to form location-specific posterior predictions. On the centered estimation grid $\{\eta_k\}$, with fitted masses $\widehat{p}_k$, define

$$\pi_{jk} \propto \widehat{p}_k \prod_{t \in \{2011,2016,2021\}} \widehat{f}_{q(j)} \left(r_{jt} - \eta_k \Delta W_{jt}\right)^{1/2}, \qquad \sum_k \pi_{jk} = 1. \tag{36}$$

The exponent tempers the combined influence of signals sharing the same 2006 baseline. This is a composite posterior calculation; the adjustment approximates dependence rather than estimating a full joint error process. The posterior mean community water value is

$$\widehat{\omega}_j^{post} = \widehat{\overline{\omega}} + \sum_k \eta_k \pi_{jk}. \tag{37}$$

Less informative responses receive greater pooling toward the population distribution. We retain the posterior distribution as well as its mean to integrate latent-value uncertainty into nonlinear counterfactuals.

Finally, we express values in AUD/ML by multiplying each community's utility coefficient by its monetary-conversion factor and dividing by the absolute estimated coefficient on log housing costs, as detailed in Appendix Section F. In Appendix Section H we show how we propagate estimation uncertainty using paired migration-count and location-weight bootstrap replications, re-estimating the distribution and retaining the corresponding price coefficient in each replication.

[15] In Appendix Section D.5 we show that among a family of distributional restrictions from normal to more flexible, the two-Gaussian mixture strikes the balance between sparsity and flexibility, performing best out of sample.

[16] We center the fitted distribution to recover the model's mean-zero heterogeneity. Let $\widehat{\mu} = \int \theta \, d\widehat{F}(\theta)$. We report $\widehat{\overline{\omega}} = \widehat{\overline{\omega}}^{IV} + \widehat{\mu}$ and define $\eta = \theta - \widehat{\mu}$. The corresponding centered signals are $r_{jt} = \widetilde{r}_{jt} - \widehat{\mu}\Delta W_{jt}$. Holding the fitted noise densities fixed, this reparameterization preserves the likelihood and total community water values.

6.2 Productive Value of Water

We construct agricultural producers' marginal values of water by adapting the production-function estimates of Rafey (2023) to our regional data. Rafey estimates crop-specific production technologies using detailed MDB farm microdata. We take the estimated technological parameters as given rather than re-estimating the agricultural production function.

Following Rafey, crop c production in region j is

$$Y_{jct} = \exp\{\omega_{jct}\} \left[\alpha_c (W_{jct} + \theta_c R_{jct})^{\frac{\sigma_c - 1}{\sigma_c}} + (1 - \alpha_c) K_{jct}^{\frac{\sigma_c - 1}{\sigma_c}}\right]^{\frac{\sigma_c}{\sigma_c - 1}\beta_{cW}} L_{jct}^{\beta_{jcL}}. \tag{38}$$

We take $(\widehat{\alpha}_c, \widehat{\theta}_c, \widehat{\sigma}_c, \widehat{\beta}_{cW}, \widehat{\beta}_{cL})$ from Rafey's preferred estimates. Appendix Table A7 reports the crop-specific values used in our calculations.

Two adjustments are required to apply these farm-level estimates to our regional data. First, we allow the effective labor elasticity to differ across SA2s because aggregation across heterogeneous farms can cause regional labor responses to differ from within-farm responses. We parameterize

$$\beta_{jcL} = \widehat{\beta}_{cL} \exp\{\delta_j\}, \tag{39}$$

and recover the persistent regional adjustment δ_j from regional labor expenditures and agricultural output, shrinking noisy regional estimates toward the farm-level estimates. Second, conditional on these labor elasticities, we invert Equation (38) to recover the crop-region-year productivity shifter ω_{jct}. Appendix Section E describes both steps in detail.

We then evaluate the marginal product of irrigation water at the observed crop-region-year allocation:

$$\widehat{V}_{jct}^{P} = F_W\left(W_{jct}, R_{jct}, K_{jct}, L_{jct}; \widehat{\omega}_{jct}, \widehat{\Theta}_c\right), \tag{40}$$

where $\widehat{\Theta}_c$ collects the crop-specific production parameters. By the envelope result in Equation (14), this marginal product is the producer's marginal value of water.

For the market analysis, we require a single producer value for each region rather than separate values for each crop. We aggregate crop-specific marginal values using their observed shares of irrigation water,

$$\widehat{V}_{jt}^{P} = \sum_c \frac{W_{jct}}{\sum_{c'} W_{jc't}} \widehat{V}_{jct}^{P}. \tag{41}$$

This aggregation values a marginal proportional increase in irrigation holding fixed the observed allocation of water across crops.[17]

[17] Allowing the crop allocation itself to respond to additional water would require modeling crop choice and within-region water reallocation. Our benchmark instead holds the observed crop mix fixed.

7 Results

7.1 Residential demand parameter estimates and community value of water

We begin by reporting the residential preference estimates that underlie our measure of community water values. We estimate our demand specification with a panel of SA1-level migration data from 2006–2021 in five-year intervals. We set the discount parameter $\beta = 0.95$ and the continuation correction smoothing parameter using Jeffreys prior.[18]

First-stage estimates: Indirect flow utilities and moving cost. The first-stage estimation recovers indirect flow utility parameters, $\delta_{jt\tau}$, origin SA2 fixed-effects, $\gamma_{j_{t-1}}$, and the moving cost per kilometer, κ. The second-stage will decompose indirect utility. Our estimate of moving cost implies that each kilometer imposes a utility cost equivalent to a 2.1 percent increase in rent. At the mean rent in our sample and using the estimated price elasticity in Table 1, this corresponds to approximately AUD $247 per year for each additional kilometer moved.

Second-stage estimates: Residential preferences. In the second stage, we decompose the recovered flow utilities into observed characteristics and residual local utility. Table 1 reports the resulting preference estimates. Households prefer communities with higher incomes and greater local irrigation, while higher housing costs reduce residential utility. A one-log-point increase in household income raises flow utility by 4.41, while a one-log-point increase in rent reduces flow utility by 2.10.[19] The estimated coefficient on local irrigation is 0.15 and is precisely estimated. Households also prefer communities with a larger stock of long-term residents: the estimated coefficient on tenure stock is 1.17, although it is less precisely estimated. The instruments strongly predict the endogenous location characteristics. The first-stage F-statistics are 382 for income, 524 for rent, 16 for tenure stock, and 1,137 for irrigation.

The estimates imply economically meaningful preferences over the observed attributes of communities while leaving substantial variation in local utility unexplained by these characteristics. The target parameter from this demand estimation is the coefficient on price which we use to both back out the location value from Equation 10 and convert household willingness to pay for water into money metric utility. While the preference parameters for income, local irrigation, and tenure stock provide additional controls in our demand estimation and allow for us to explore a decomposition of household value for water, the price coefficient is the primary target of this analysis. In Appendix Table A6, we show that the price coefficient is robust to alternative demand specifications that contain fewer controls.

[18] See Appendix Section B.4 for more details.

[19] Translating our estimates into an own-price elasticity at the SA2 level, which is more comparable to US Census tract sizes, we get a median yearly elasticity of 0.254. This elasticity is in the range of estimates, for the yearly level, from Diamond (2016), Cook et al. (2025), and Kashner (2026).

Table 1: Residential Choice Parameters

Dependent Variable:	$\hat{\delta}_{jt\tau}$
log(Rent)	-2.096
	(0.528)
log(Income)	4.412
	(1.428)
Local Irrigation	0.153
	(0.042)
Tenure stock	1.169
	(0.671)
Controls	
Death Exposure	Yes
Tenure Status	Yes
SA1	Yes
Water District × Year	Yes
Fit statistics	
Observations	59,284
R^2	0.2512
F-test (1st stage), Water	1,137.1
F-test (1st stage), log(Income)	382.12
F-test (1st stage), log(Rent)	523.84
F-test (1st stage), Tenure stock	16.35

Notes: The dependent variable is flow indirect utility recovered from the first-stage residential choice model. The specification includes tenure-state, destination, and water-district-by-year fixed effects and controls for the local death rate. Income, rent, tenure stock, and irrigation are instrumented as described in Section 6. Standard errors are clustered by destination. First-stage F-statistics for the excluded instruments are reported at the bottom of the table.

7.2 Community Value of Water

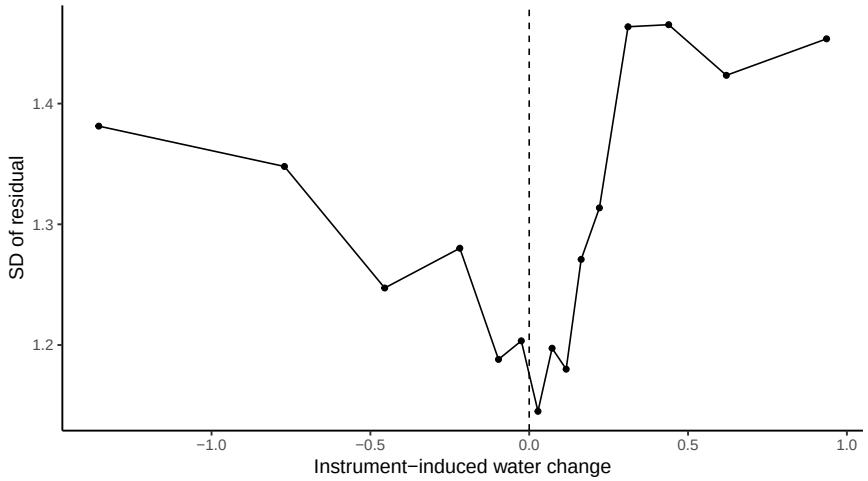
We next estimate the value communities place on retaining local irrigation. Our measure uses households' revealed residential preferences to measure the total effect of irrigation on the attributes of a community that residents value without estimating each channel separately.

First, we estimate the stacked control function regression specified in Equation 30 to recover the mean community willingness to pay for local irrigation. We find that a mean of \$141/ML, with a 95% bootstrap confidence interval of \$106–\$186/ML. This mean value, $\widehat{\overline{\omega}}$, captures the average effect of irrigation on community utility, including both its conditional value and its effects on other valued attributes of place.

Second, we recover the heterogeneity of values across locations. Figure 4 provides descriptive

evidence of this heterogeneity by plotting the dispersion of residual utility changes, r_{jt}, from Equation (31) against instrument-induced water reallocation. Residual utility changes are relatively concentrated among communities experiencing little reallocation and become more dispersed as the magnitude of water changes increases. This pattern is consistent with persistent differences across communities in the utility effect of irrigation.

Figure 4: Heterogeneity in the Effect of Water Reallocation on Local Utility



Notes: The figure plots the standard deviation of residual changes in local utility against instrument-induced changes in irrigation. Residual utility changes remove the average effect of water reallocation. Points report standard deviations within bins of predicted water changes.

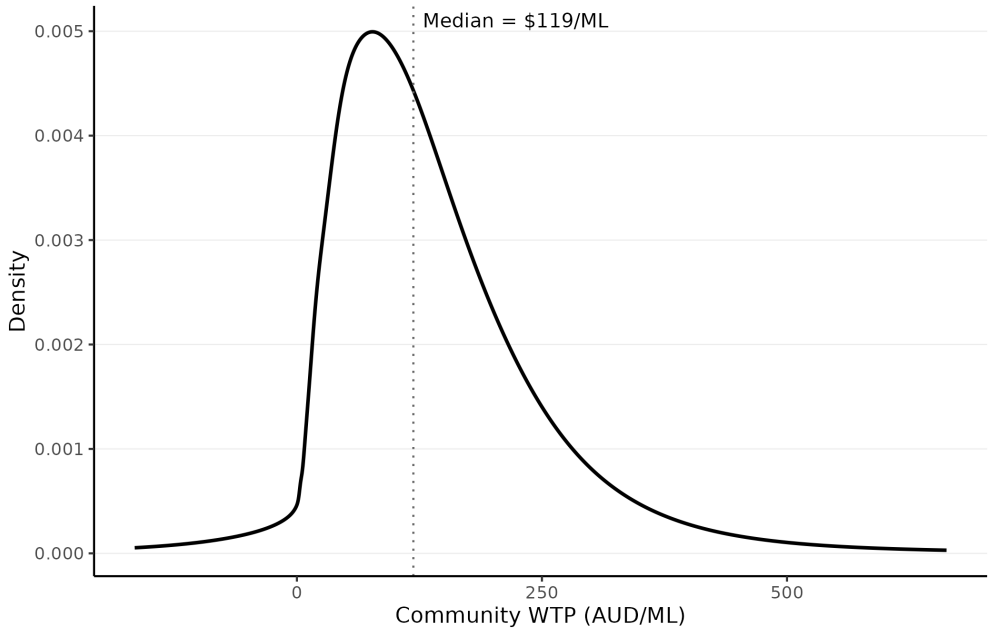
Our baseline two-Gaussian specification exhibits a thick right tail with an estimated median of \$119/ML (95% confidence interval: \$87–\$160/ML).[20] Community values are substantially heterogeneous. The estimated population interquartile range is \$67–\$191/ML, and the 10th and 90th percentiles are \$33 and \$280/ML. Figure 5 displays the fitted population distribution after applying the location-specific monetary conversions described in Appendix F. The median value is approximately half the average market price of \$235/ML over the study period. An estimated 15.8% of communities value water above this benchmark, with a 95% confidence interval of 8.1–28.5%.

The confidence intervals propagate uncertainty in recovered residential utilities, housing-cost preferences, and the distribution of community water values through the paired migration-count and location-weight bootstrap (Appendix H). Local monetary-conversion factors are held fixed.

We also combine the fitted population distribution with each community's observed utility responses to recover its posterior distribution of water values. These distributions summarize both location-specific values and the remaining uncertainty about them, with less informative responses receiving greater pooling toward the population distribution. Appendix Figure A8 reports the distribution of posterior means. In the counterfactual analysis below, we integrate over the location-specific posterior distributions to account for uncertainty about community values when evaluating

[20] Appendix Figure A6, we report summary statistics for estimated distributions under normal and non-parametric specifications. Appendix Figure A7 depicts the densities. While the estimated variance increases moving from normal to non-parametric, the results are not meaningfully different.

Figure 5: Distribution of Community Values



Notes: The figure plots the estimated population distribution of community willingness to pay for local irrigation across SA1s. Values are expressed in 2021 AUD per ML.

compensation and consent.

7.3 Agricultural production and the private value of water

We apply our adapted agricultural production model to a panel of SA2-level data from 2005–2020 in five year intervals. As described in Section 6, we apply the crop-specific production technology estimates from Rafey (2023) to our regional production data. Because these parameters are estimated at the farm level, we allow the labor elasticity to adjust across SA2s when applying the production function to regional aggregates. The mean estimated adjustment, $\exp\{\hat{\delta}_j\}$, is 0.73, with an interquartile range of 0.36–1.27. Appendix Table A9 reports the distribution of these adjustments and the resulting productivity estimates.

We use the estimated production functions to calculate the marginal product of irrigation water for each crop, SA2, and year. We then aggregate across crops within an SA2-year using observed shares of applied irrigation water. This produces the location-year productive value V_{jt}^{P} used in the market-design counterfactuals. The resulting distribution is highly dispersed and right-skewed. Across SA2-years, the median productive value of water is \$265/ML, with an interquartile range of \$137–\$840/ML. The mean is substantially higher, at \$934/ML, reflecting a long right tail of locations in which the marginal product of water is particularly high.

Much of this dispersion reflects differences across agricultural uses. Median marginal values are \$156/ML for dairy and pasture, \$180/ML for annual crops, and \$1,148/ML for perennial crops. Appendix Table A10 and Appendix Figure A9 report the full crop-specific distributions. Appendix Figure A10 shows that productive values also vary over time with hydrological conditions, rising in

dry years and falling in wetter years. For the counterfactual analysis below, we retain the location and year specific productive values rather than collapsing this variation to a single value for each region.

7.4 Community vs. private value for water

The median community value for local irrigation is \$119/ML, compared with a median productive value of \$265/ML across SA2-years. The distributions have substantial overlapping support: the interquartile range of community values is \$67–\$191/ML, while that of productive values is \$137–\$840/ML. In Appendix Figure A11, we depict the distribution of productive gains and community values. Community values can be substantial relative to the private values that govern the allocation of water.

We next examine whether communities place greater value on retaining irrigation in locations where water generates higher agricultural returns. We match community willingness to pay to the productive value of water and induced labor spending per ML in the corresponding SA2 and year. We calculate model-based correlations using the posterior distribution of community values, incorporating the remaining posterior variance rather than treating posterior means as known values.

Productive and community values are weakly positively correlated. Pooling across years, the estimated correlation is 0.040, with a 95% bootstrap interval of [0.017, 0.061]. Although distinguishable from zero, the relationship is small. Locations with high agricultural returns to water do not consistently exhibit high community values.

We also correlate community values with the agricultural model implied marginal increase in labor spending from additional irrigation. The correlation is about double that of private farmer profit, though still small. The pooled correlation is 0.082 [0.051, 0.114]. These results indicate that agricultural labor spending provides more signal about where communities value water, though along with private profit, both provide limited information about community willingness to pay to retain irrigation.

We next ask how farmer and comunity values relate to the direction of water reallocation following market integration. Table 2 relates post-reform changes in local irrigation to productive and community values. Columns 1–3 use the log ratio of post-reform to pre-reform irrigation, while Columns 4–6 consider whether a location experienced a net increase in irrigation.

Water reallocation is strongly associated with productive values. In the joint specification, a 10% higher productive value is associated with approximately 1.3% more irrigation relative to the pre-reform period. In contrast, a 10% higher community value is associated with approximately 0.8% less irrigation, although this relationship is less precisely estimated. The logit specifications exhibit the same pattern: productive values are positively associated with the probability that a location gained water, while community values are negatively associated with that probability.

These results show that productive and community values represent distinct dimensions of the value of local water. Market integration moved water toward locations where its productive value was higher, but productive values contain little information about where the community value of

Table 2: Water Reallocation and the Value of Water

Outcome	Log water ratio			Water gain		
Model:	(1) OLS	(2) OLS	(3) OLS	(4) Logit	(5) Logit	(6) Logit
Log farmer value	0.122		0.127	0.209		0.222
	(0.029)		(0.029)	(0.065)		(0.065)
Log community value		-0.055	-0.082		-0.197	-0.238
		(0.048)	(0.048)		(0.087)	(0.089)

Notes: Columns 1-3 regress the log ratio of post to pre-period water availability on estimated productive and community values of water. Columns 4-6 use an indicator for an increase in local water availability as the dependent variable. All specifications include year fixed effects. Standard errors are clustered at the SA2 level.

water is high. Whether the private gains generated by a transfer are sufficient to compensate the communities it affects therefore depends on the joint distribution of buyer and seller productive and community values. We turn to this question in the next section.

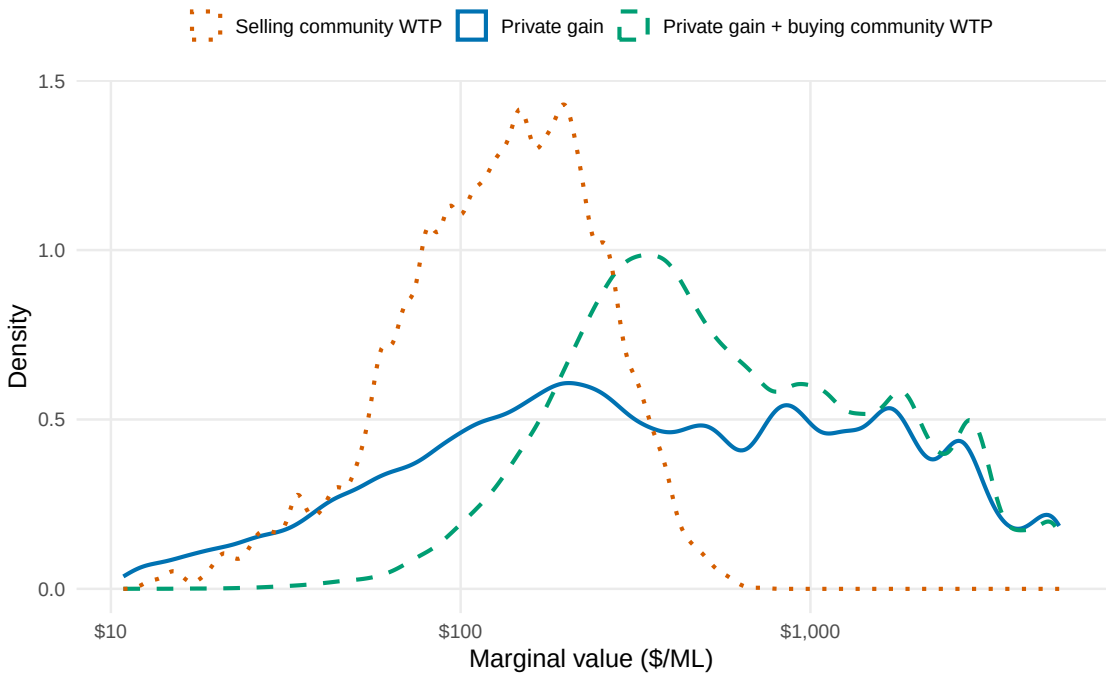
8 Counterfactuals: Local Control and Market Reallocation

We now use the estimated community and productive values to quantify the market-design tradeoffs introduced in Section 2 about how community consent changes the implementation of privately gainful transfers. Consider a potential transfer of one ML of water from a seller s to a buyer b. The private gain from the transfer is $G^P_{sb} = V^P_b - V^P_s$. In the market benchmark, the buyer and seller can trade whenever $G^P_{sb} > 0$, without obtaining the consent of either community. We normalize the set of these privately gainful trade opportunities to one and ask how many continue to occur when selling communities are given control over whether water can leave.

As in Section 2, the consequences of community veto rights depend on two features of the environment. The first is the surplus available to secure community consent. The buyer and seller can use the private gain G^P_{sb}; if the benefit to the buying community can also be mobilized, available surplus rises to $G^P_{sb} + V^C_b$. The second is information about the selling community's value. When V^C_s is known, compensation can be tailored to the amount required to secure consent. When it is private information, traders must make an offer without knowing the community's reservation value.

Figure 6 illustrates the magnitudes underlying these comparisons. The dotted distribution reports the compensation required to secure selling-community consent, V^C_s, while the solid distribution reports the private gains available from trade, G^P_{sb}. There is substantial overlap: many privately gainful trades generate enough surplus to compensate the selling community, while others do not. The dashed distribution adds the benefit to the buying community, V^C_b, shifting available

Figure 6: Comparing trade surplus to community values for water



Notes: The figure compares the distributions of community and private values associated with potential water transfers. Selling community value is the selling community's willingness to pay to retain one ML of water locally. Private gain is the difference between the buying and selling farmers' productive values of one ML of water, $G^P_{sb} = V^P_b - V^P_s$. Private gain + buying community value adds the buying community's willingness to pay for the local benefits of receiving the water, V^C_b, to the private gain from trade. The horizontal axis is shown on a logarithmic scale and values are expressed in AUD per ML.

surplus to the right. All three distributions are highly dispersed, making both their joint realization and the information available about community values consequential for which trades proceed. The marginal distributions alone do not determine market performance. Whether a particular trade can secure community consent depends on the joint realization of V^P_s, V^P_b, V^C_s, and V^C_b. We construct this joint distribution by simulating potential bilateral trades.

We construct random bilateral markets separately within each year. Locations are paired sequentially without replacement within a market, with the higher-productive-value location designated as the buyer. We repeat the matching exercise across random markets and retain the same opportunities across mechanisms and bootstrap replications. This provides a common benchmark for comparing trading rules without estimating an equilibrium matching process.[21] It does not imply that the simulated pairs reproduce the transactions observed in the MDB, but rather represents the degree to which community consent creates trade frictions given the estimated joint distribution of private gains and community values.

For each fitted set of parameters, we draw a value for each community from its local posterior and evaluate every mechanism on that realization. A community's draw is shared across all its

[21] Observed matching in the MDB reflects heterogeneous trading rules, institutional constraints, and transaction costs that are difficult to model without imposing substantial additional structure (Rafey, 2023).

appearances in the simulated markets and years. We average outcomes across posterior realizations, retaining latent uncertainty in participation and compensation decisions. Under full information, the mechanism observes the realized community value; under private information, it observes only the information specified below. The evaluator's use of local posteriors does not give traders access to those posteriors.

Reported point estimates average across random markets and years. Confidence intervals are the 2.5th and 97.5th percentiles of posterior-averaged statistics across the paired residential-choice and location-weight bootstrap replications. The matching, farmer values, and local monetary-conversion factors are held fixed. Appendix H describes the resampling procedure and within-market weighting.

Community Veto under Full Information We first grant selling communities veto rights and suppose that all values are observed. If compensation can be financed only from the private gains generated by the trade, a transfer proceeds whenever

$$G_{sb}^{P} \geq V_{s}^{C}. \tag{42}$$

Full information isolates a surplus constraint: trades that fail this condition do not generate sufficient private gains to make the selling community whole.

Table 3 reports the results. With private surplus alone, 73.3% of privately gainful trade opportunities continue to occur when selling communities receive veto rights (95% confidence interval: 68.0–78.3%). Selling communities receive average compensation of \$97.7/ML among implemented trades. The remaining 26.7% cannot generate sufficient private surplus to compensate the selling community; these unrealized trades have average private gains of \$58.8/ML.

The result changes substantially when benefits to the buying community can also be mobilized. A trade can then secure selling-community consent whenever

$$G_{sb}^{P} + V_{b}^{C} \geq V_{s}^{C}. \tag{43}$$

Under this condition, 89.5% (95% confidence interval: 87.8–91.0%), an increase of 16.2 percentage points. Average compensation is \$101.0/ML, and the mean private gain among unrealized opportunities is \$53.1/ML. Mobilizing benefits to the receiving community preserves a substantial share of opportunities that fail the private-financing constraint.

Community Veto under Private Information Full information abstracts from a second friction emphasized in Section 2: the compensation required to secure a community's consent may itself be private information. We suppose that the selling community observes their draw of V_{s}^{C} from the location specific posterior distribution, while traders know only the population distribution and the communities money-metric conversion factor. For each potential trade, the buyer and seller make a take-it-or-leave-it compensation offer. A higher offer increases the probability that the community consents but reduces the surplus retained by the traders when the offer is accepted. We allow traders

Table 3: Summary of bilateral trading protocols

Market design	Buying surplus	Trades (%) implemented	Avg. Payment ($/ML)	Avg. lost gain ($/ML)
No compensation	–	100.0	0.0	–
Full information	No	73.3	97.7	58.8
		[68.0, 78.3]	[74.2, 127.5]	[49.0, 70.9]
Full information	Yes	89.5	101.0	53.1
		[87.8, 91.0]	[77.0, 131.8]	[45.8, 61.9]
Trader TIOLI	No	61.9	162.6	158.8
		[57.3, 66.2]	[132.7, 197.4]	[141.0, 177.5]
Trader TIOLI	Yes	76.6	162.3	198.4
		[74.7, 78.8]	[132.7, 197.4]	[172.0, 223.8]

Notes: The table reports outcomes under alternative bilateral trading protocols. Trade opportunities are transfers with positive private gains. "Buying surplus" indicates whether the buying community's value can also finance compensation. "Trades implemented" is the share of trade opportunities that proceed. "Avg. payment" is mean compensation to selling communities among implemented trades, and "Avg. lost gain" is the mean private gain among unrealized trades. Under full information, selling-community values are observed and compensation is tailored to secure consent. Under Trader TIOLI, these values are privately known and traders make an optimal take-it-or-leave-it offer. Monetary values are in AUD per ML.

to choose the offer that maximizes their expected surplus.[22]

Private information substantially reduces the amount of trade that survives community veto. When compensation is financed only from private gains, the share of trade opportunities implemented falls from 73.3% under full information to 61.9% under the Trader TIOLI mechanism. Average compensation among implemented trades rises from $97.7/ML to $162.6/ML. Communities whose values lie below the optimal offer receive information rents, while some communities whose values could have been compensated from the available surplus reject offers below their reservation values.

The forgone trades illustrate the cost of this information friction. Trades that fail under full information with private surplus have average private gains of only $58.8/ML. Under private information, unrealized trades have average private gains of $158.8/ML. Thus, private information prevents some trades with substantial productive gains even though sufficient surplus could, under full information, have been used to secure community consent.

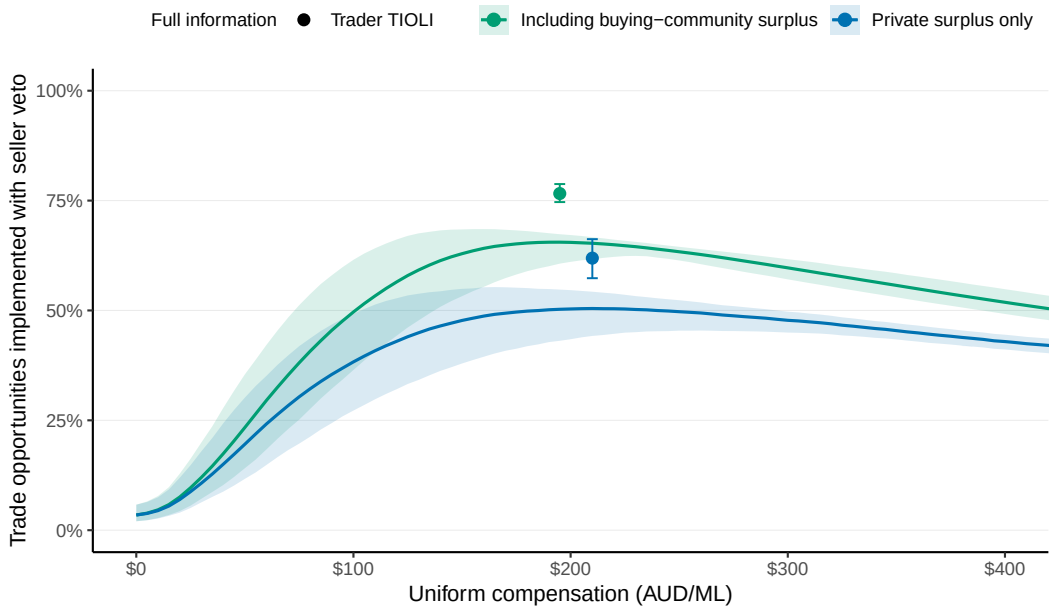
Allowing buying-community contributions raises TIOLI implementation to 76.6% (95% confidence interval: 74.7–78.8%), compared with 89.5% under full information with the same financing. Average payment is $162.3/ML, and mean private gain among unrealized opportunities is $198.4/ML. Additional financing relaxes the surplus constraint, but private information continues

[22] Appendix Section G provides the details of the mechanism. The optimal offer is the procurement analogue of a monopolist's optimal posted price (Myerson, 1981, Bulow & Roberts, 1989).

to substantially limit trade.

Uniform Compensation We also consider a uniform compensation rule under which communities retain the right to reject transfers. This benchmark separates consent from discretionary bargaining over payments, motivated by legal restrictions on conditioning land-use approval on negotiated contributions. For this alternative market institution, we require the same compensation $c \geq 0$ for every implemented transfer, while retaining the selling community's veto. A transfer proceeds if compensation, $c \geq V_s^C$, is greater than the seller community's realized willingness to accept and compensation is less than the available surplus from the trade. We consider both when compensation is financed by private gains and buy the augmented surplus from buying community spillovers. The primary tension for this market mechanism is that increasing the payment can secure consent from more communities, but makes compensation unaffordable for more trades.

Figure 7: Uniform Compensation with Community Veto



Notes: Curves report the fraction of privately gainful opportunities satisfying $C_s \leq c \leq S_{sb}$ at each payment c, with private financing or buying-community contributions. Shading shows pointwise 95% bootstrap percentile intervals. Trader TIOLI markers show its implementation rate at the evaluated uniform price with the closest implementation rate; the horizontal coordinate is not the trader's offer. A marker above the uniform curve's maximum has no equivalent uniform payment. Marker error bars are 95% bootstrap intervals for TIOLI implementation.

Figure 7 plots the share of trades that can be implemented as a function of the uniform payment.[23] Among the evaluated prices, the trade-maximizing payment is \$210/ML under private financing and \$195/ML with buying-community contributions, implementing 50.4% and 64.5% of

[23] Appendix Figure A12 considers uniform compensation without community veto, reporting both the share of trades implemented and the share of selling communities whose losses exceed their compensation. The share of implemented trades increases because community consent is not required, but many communities are left undercompensated and implementation shares still fall behind the TIOLI market mechanism.

opportunities, respectively. The corresponding 95% confidence intervals for implementation are [44.2, 54.3] and [60.6, 67.4]. Even at their baseline-optimal levels, uniform payments implement fewer trades than trader TIOLI offers, which implement 61.9% and 76.6% under the respective financing arrangements. The gaps of 11.5 and 11.1 percentage points illustrate the additional trade enabled by tailoring compensation offers to individual opportunities.

These counterfactuals show that the effect of community control on market reallocation depends on more than the magnitude of local losses. Under full information, most privately gainful trades can still proceed while fully compensating selling communities, particularly when the benefits accruing to buying communities can also be mobilized. Private information creates an additional friction: compensation must leave information rents to some communities, while other trades fail even when their gains would have been sufficient to secure consent under full information. Uniform compensation cannot target payments to heterogeneous community values and preserves less. These results highlight the central market-design tradeoff created by local control: the extent to which markets can accommodate the interests of affected nonparticipants depends jointly on the surplus generated by reallocation, who can contribute that surplus, and the information available to direct compensation where it is most valuable.

9 Conclusion

Market integration creates gains by reallocating resources toward higher-value uses, but the same reallocation can impose substantial consequences on people outside the transaction. When affected communities have the power to block trade, these local consequences become part of the market-design problem. We study this tension in Australia's Murray–Darling Basin, where water-market integration generated large productive gains while moving irrigation activity across rural communities.

We find that communities place substantial and highly heterogeneous value on retaining irrigation. These values are only weakly related to the productive value of water, so the market reveals where gains from reallocation can be created but provides little information about where compensation is most valuable. This distinction matters when communities receive control over trade. Under full information, many profitable trades can still proceed while fully compensating affected communities, particularly when benefits accruing to buying communities can also be mobilized. Private information makes local control more restrictive because compensation cannot be perfectly targeted. Uniform compensation avoids cannot target heterogeneous community values and therefore underperforms our bargaining protocol.

The broader lesson is that local control need not simply be understood as a barrier to efficient markets. Giving affected nonparticipants rights over transactions brings values that markets otherwise ignore into the bargain. How costly those rights are for market reallocation depends on whether the gains from trade are large enough to accommodate those values, whether gains accruing outside the transaction can be mobilized, and whether compensation can be targeted when local

values are privately known. These forces arise whenever market integration reallocates economic activity across place or use. Understanding them can help distinguish settings in which local control primarily blocks gains from trade from those in which bargaining can preserve reallocation while giving affected communities a meaningful role in whether it occurs.

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A Measuring local water use for irrigation

Our empirical analysis requires a measure of irrigation water availability at the SA1 level. Administrative records on water allocations and trades are not available at this spatial resolution. We therefore construct a measure using satellite estimates of actual evapotranspiration (ET). This appendix describes the construction of the measure and validates it against administrative measures of water allocations and applied irrigation water.

Residual evapotranspiration. Actual ET reflects both irrigation and naturally available water. To avoid mechanically attributing evapotranspiration from rivers and other surface waterways to irrigation, our preferred measure excludes pixels within 500 meters of major waterways. We also construct the measure using exclusion bands of 0, 100, and 250 meters and find that our results are robust to these alternative definitions. Actual ET also responds to rainfall and atmospheric conditions. We isolate variation in ET unexplained by these meteorological conditions.

For each SA1 j, calendar month m, and year t, we estimate

$$ET_{jmt} = \rho_0 Rain_{jmt} + \rho_1 Rain_{jm,t-1} + \psi PET_{jmt} + \gamma_{jm} + \varepsilon_{jmt}, \tag{44}$$

where ET_{jmt} is actual evapotranspiration per hectare, $Rain_{jmt}$ is contemporaneous rainfall, $Rain_{jm,t-1}$ is lagged rainfall, and PET_{jmt} is potential evapotranspiration. The SA1-by-calendar-month fixed effects γ_{jm} absorb persistent local differences in climate, vegetation, and seasonality. We define residual ET as

$$\widehat{\varepsilon}_{jmt} = ET_{jmt} - \widehat{ET}_{jmt}. \tag{45}$$

Positive residual ET therefore indicates that observed evapotranspiration is greater than predicted by local weather conditions and persistent seasonal differences.

Annual water availability. We construct our preferred annual measure using residual ET during the dry months, October through March. Irrigation should contribute more strongly to ET during these months because naturally available water is relatively scarce. For SA1 j in year t, we define

$$W_{jt} = \sum_{m \in \mathcal{M}^{Dry}} \widehat{\varepsilon}_{jmt}, \qquad \mathcal{M}^{Dry} = \{\text{October}, \ldots, \text{March}\}. \tag{46}$$

Thus, W_{jt} measures annual evapotranspiration in excess of that predicted by weather during the period in which irrigation is most likely to be responsible for this excess water use. Because ET is measured per unit of land, W_{jt} is an intensive-margin measure of local irrigation activity.

For the event-study analysis, we normalize water so that one unit corresponds to 0.096 ML/ha of residual ET. This is approximately a 10 percent change in irrigation intensity at the average SA1. The event-study coefficients can therefore be interpreted as the effect of approximately a 10 percent change in local irrigation activity.

Table A1: Administrative Water Allocations and Residual Evapotranspiration

Residual ET Model:	Annual (1)	Dry (2)	Wet (3)
Variables			
Total allocation	0.7312*	0.6511*	-0.0279
	(0.3757)	(0.3286)	(0.1372)
Fixed-effects			
Water District	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Observations	440	440	440
R^2	0.31945	0.78145	0.93048

Notes: The dependent variables are annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. The explanatory variable is total administrative water allocation. All specifications include water-district and year fixed effects. Standard errors are clustered at the water-district level.

Water exposure. For specifications that require a single measure of the reallocation induced by the water-market reform, we compare average water availability before and after 2007. Specifically, we define

$$\Delta W_{j,t} = \overline{W}_{j,\text{post}\leq t} - \overline{W}_{j,\text{pre}}, \tag{47}$$

where $\overline{W}_{j,\text{pre}}$ and $\overline{W}_{j,\text{post}\leq t}$ denote average dry-season residual ET before and after the 2007 reform up to year t, respectively. Positive values of $\Delta W_{j,t}$ correspond to locations in which irrigation activity increased relative to the pre-reform period, while negative values correspond to locations in which it declined.

Validation with administrative water use. We validate our satellite-based measure using administrative records of annual water allocations at the water-district level. Because these data directly measure the amount of water made available to users, but are observed at a substantially coarser geography than our SA1 analysis, they provide a useful external validation of the residual ET measure.

We aggregate residual ET to the corresponding water-district and estimate

$$W_{rt}^{ET} = \beta A_{rt} + \alpha_r + \lambda_t + u_{rt}, \tag{48}$$

where W_{rt}^{ET} is residual ET in water-district r, A_{rt} is total administrative water allocation, and α_r and λ_t are region and year fixed effects. Standard errors are clustered by water-district.

Table A1 compares residual ET constructed over the full year, the dry season, and the wet season. Administrative allocations are positively related to both annual and dry-season residual ET,

with a particularly strong relationship during the dry season. A one-unit increase in total allocation is associated with a 0.65-unit increase in dry-season residual ET. In contrast, allocations have essentially no relationship with wet-season residual ET: the estimated coefficient is −0.03. This seasonal contrast supports our interpretation that residual ET during the dry season primarily captures irrigation water rather than unobserved natural variation in evapotranspiration, and motivates our preferred October–March measure.

We further decompose total water availability into direct allocations, environmental buybacks, and net agricultural transfers. Appendix Table A11 reports the results. The positive relationship with dry-season residual ET is driven primarily by direct water allocations. Environmental buybacks and net agricultural transfers are considerably noisier and are estimated imprecisely at this level of aggregation. As in the aggregate specification, none of these components meaningfully predicts wet-season residual ET.

Finally, we conduct a similar validation using ABS measures of applied irrigation water at the finer SA2 level. We view these data as a noisier measure of annual local water availability, but they allow validation closer to the spatial scale of our empirical analysis. The results are similar: dry-season residual ET is positively associated with observed applied irrigation water, whereas wet-season residual ET is not. Appendix Table A12 reports these results.

B Derivation of the Dynamic Residential Choice Estimator

This appendix derives the first-stage residential choice estimator described in Section 6. We first use the Hotz–Miller inversion to express current relative choice probabilities in terms of flow utility, moving costs, next-period conditional choice probabilities, and continuation values. We then exploit the renewal structure of the outside option to eliminate the remaining continuation values and obtain the estimating equation used in the first-stage likelihood.

B.1 Dynamic choice probabilities

A household enters period t with individual state

$$x_{t-1} = (j_{t-1}, \tau_{t-1}), \tag{49}$$

where j_{t-1} denotes the household's previous location and τ_{t-1} its tenure in that location. The household chooses destination j and receives flow utility

$$u_{ht}(j \mid x_{t-1}) = \overline{u}_{jt\tau} + MC_t(j, j_{t-1}) + \varepsilon_{hjt}, \tag{50}$$

where $\overline{u}_{jt\tau}$ is mean flow utility from choosing destination j in period t and entering tenure state τ, $MC_t(j, j_{t-1})$ captures the utility cost of moving from the household's previous location, and ε_{hjt} is an idiosyncratic Type-I extreme value preference shock. We suppress aggregate states throughout the derivation for notational convenience.

Under the conditional-independence assumptions described in the main text, define the expected value of entering period t in state x_{t-1}, prior to the realization of the idiosyncratic shocks, as

$$\overline{V}_t(x_{t-1}) = \log\left[\sum_{j\in\mathcal{J}} \exp\left\{v_t(j, x_{t-1})\right\}\right], \tag{51}$$

where the choice-specific value is

$$v_t(j, x_{t-1}) = \overline{u}_{jt\tau} + MC_t(j, j_{t-1}) + \beta\overline{V}_{t+1}(x'_j), \tag{52}$$

and

$$x'_j \equiv (j, \tau_t(j, \tau_{t-1})) \tag{53}$$

is the household state induced by choosing j.

The corresponding conditional choice probability is

$$\mathbb{P}_t(j \mid x_{t-1}) = \frac{\exp\{v_t(j, x_{t-1})\}}{\sum_{\ell\in\mathcal{J}} \exp\{v_t(\ell, x_{t-1})\}}. \tag{54}$$

We normalize the flow utility of the composite outside option j_0 to zero,

$$\overline{u}_{j_0 t} = 0. \tag{55}$$

Taking the log odds of choosing an inside location j relative to the outside option gives

$$\begin{aligned}\log\frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} = \overline{u}_{jt\tau} + MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ + \beta\left[\overline{V}_{t+1}(x'_j) - \overline{V}_{t+1}(x'_{j_0})\right].\end{aligned} \tag{56}$$

The remaining difference in continuation values prevents direct estimation.

B.2 Hotz–Miller inversion and renewal

For any state x and reference action a, the Hotz–Miller inversion implies

$$\overline{V}_{t+1}(x) = v_{t+1}(a, x) - \log\mathbb{P}_{t+1}(a \mid x). \tag{57}$$

We apply this identity to each continuation value in Equation (56), using the outside option j_0 as the reference action:

$$\begin{aligned}\overline{V}_{t+1}(x'_j) - \overline{V}_{t+1}(x'_{j_0}) = v_{t+1}(j_0, x'_j) - v_{t+1}(j_0, x'_{j_0}) \\ - \log\frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}.\end{aligned} \tag{58}$$

Because the outside option has normalized flow utility, the difference in choice-specific values is

$$v_{t+1}(j_0, x'_j) - v_{t+1}(j_0, x'_{j_0}) = MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0)$$

$$+ \beta \left[\overline{V}_{t+2}(x''_{j,j_0}) - \overline{V}_{t+2}(x''_{j_0,j_0}) \right]. \tag{59}$$

The outside option provides a renewal action that eliminates the final continuation-value difference. Tenure is a payoff-relevant individual state only within the modeled MDB locations. Choosing the composite outside option resets tenure:

$$\tau_t(j_0, \tau) = 0 \qquad \forall \tau. \tag{60}$$

Thus, regardless of whether the household chooses j or j_0 in period t, choosing j_0 in period $t+1$ produces the same state in period $t+2$:

$$x''_{j,j_0} = x''_{j_0,j_0} = (j_0, 0). \tag{61}$$

It follows that

$$\overline{V}_{t+2}(x''_{j,j_0}) - \overline{V}_{t+2}(x''_{j_0,j_0}) = 0. \tag{62}$$

The outside option therefore generates finite dependence: after one additional outside choice, the states induced by the two current choices coincide. This eliminates continuation values from period $t+2$ onward without requiring us to solve the household value function.

Substituting into Equation (56) gives

$$\begin{aligned} \log \frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} = \overline{u}_{jt\tau} &+ MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ &+ \beta \left[MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0) \right] \\ &- \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}. \end{aligned} \tag{63}$$

Equation (63) is the key estimating relationship. The dynamic component of the household problem is summarized by next-period exit probabilities, while all continuation values have been eliminated.

B.3 Moving costs and first-stage flow utility

We next map Equation (63) into the parameterization used in estimation. Normalize the utility cost of remaining in the outside option,

$$MC_{t+1}(j_0, j_0) = 0. \tag{64}$$

The remaining moving-cost terms are

$$\underbrace{MC_t(j, j_{t-1})}_{\text{origin–destination}} - \underbrace{MC_t(j_0, j_{t-1})}_{\text{origin}} + \underbrace{\beta MC_{t+1}(j_0, j)}_{\text{destination}}. \tag{65}$$

The first term varies jointly across origins and destinations. Rather than introducing unrestricted origin-by-destination parameters, which would absorb much of the variation used to identify

destination preferences, we parameterize this component using distance:

$$MC_t(j, j_{t-1}) = \kappa \, \text{Dist}(j, j_{t-1}), \tag{66}$$

where the sign of κ follows our utility convention.

The second term depends only on the household's origin and is absorbed by origin fixed effects, $\gamma_{j_{t-1}}$. The third depends only on the destination and is not separately identified from destination flow utility in the first stage. We therefore absorb it into the destination-period-tenure utility $\delta_{jt\tau}$. Because this component is time invariant, it is subsequently absorbed by destination fixed effects in the second-stage decomposition of $\widehat{\delta}_{jt\tau}$.

The first-stage choice index can therefore be written as

$$\delta_{jt\tau} + \gamma_{j_{t-1}} + \kappa \, \text{Dist}(j, j_{t-1}) - \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}. \tag{67}$$

The first-stage likelihood estimates the destination-period-tenure utilities $\delta_{jt\tau}$, origin fixed effects $\gamma_{j_{t-1}}$, and distance parameter κ, taking the estimated next-period exit probabilities as given. The recovered $\widehat{\delta}_{jt\tau}$ are the tenure-specific flow utilities that enter the second-stage decomposition described in Section 6.1.

B.4 Regularization of exit conditional choice probabilities

The preceding estimator requires empirical CCPs only for the exit probabilities entering Equation (67). Because these probabilities enter logarithmically, a raw empirical exit probability of zero would make the continuation correction undefined.

Let $s = (k, t, x)$ index a state cell, let N_s denote the number of households making a choice from that state, and let N_s^0 denote the number choosing the outside option. The raw empirical exit probability is

$$\widetilde{q}_s^0 = \frac{N_s^0}{N_s}. \tag{68}$$

We regularize these probabilities using a Beta–Binomial model,

$$q_s^0 \sim \text{Beta}(\alpha_q, \beta_q), \tag{69}$$

$$N_s^0 \mid N_s, q_s^0 \sim \text{Binomial}(N_s, q_s^0). \tag{70}$$

The resulting posterior is

$$q_s^0 \mid N_s^0, N_s \sim \text{Beta}\left(N_s^0 + \alpha_q,\ N_s - N_s^0 + \beta_q\right), \tag{71}$$

and we use its posterior mean,

$$\widehat{q}_s^0 = \frac{N_s^0 + \alpha_q}{N_s + \alpha_q + \beta_q}. \tag{72}$$

Equivalently,

$$\widehat{q}_s^0 = \frac{N_s}{N_s + \alpha_q + \beta_q}\widetilde{q}_s^0 + \frac{\alpha_q + \beta_q}{N_s + \alpha_q + \beta_q}\frac{\alpha_q}{\alpha_q + \beta_q}. \tag{73}$$

The regularized estimate therefore shrinks the raw empirical exit probability toward the prior mean $\alpha_q/(\alpha_q + \beta_q)$, with the amount of shrinkage declining in the number of observations in the state cell. We set $(\alpha_q, \beta_q) = (1/2, 1/2)$ in the baseline specification.

This regularization is required only for the exit CCPs used to construct the continuation correction. The likelihood in Equation (26) is evaluated directly from observed destination counts. A destination with zero observed choices from a particular state therefore contributes zero to the corresponding likelihood term and does not require the construction or smoothing of a positive empirical destination CCP.

C Instruments for Residential Location Characteristics

This section describes the instruments used for household income, housing costs, and the local stock of long-term residents in the second-stage residential-choice estimation.

C.1 Industry shift-share instrument for income

We construct a shift-share measure of predicted local employment growth using each SA1's industry composition in 2001. We first map employment by industry from 2001 Census Districts to SA1s and calculate the share of SA1 j's employment in industry c,

$$s_{jc,2001} = \frac{L_{jc,2001}}{\sum_{c'} L_{jc',2001}}.$$

For each subsequent Census year $t \in \{2006, 2011, 2016, 2021\}$, we calculate the growth of national employment in the corresponding industry relative to 2001,

$$g_{ct} = \frac{L_{ct}^{\text{AUS}} - L_{c,2001}^{\text{AUS}}}{L_{c,2001}^{\text{AUS}}}.$$

The instrument used in our baseline specification is

$$Z_{jt}^I = \sum_c s_{jc,2001} g_{ct}.$$

It predicts differential changes in local labor-market conditions from predetermined industry composition and subsequent national industry growth. The national growth rates in this baseline construction include employment in the focal SA1.

C.2 Predicted retention instrument for long-term residence

We instrument the local stock of long-term residents with predicted retention based on the SA1's lagged age-by-tenure composition and retention patterns outside its own SA2. For each SA1 j, age

group a, and Census year t, let $s^{1\text{–}4}_{ja,t-5}$ and $s^{5+}_{ja,t-5}$ be the shares of the entire lagged SA1 population in the corresponding age-by-tenure cells. These shares are constructed from the preceding Census and shifted forward five years to align them with the outcome year.

For each age group and year, we calculate the shares of individuals observed in the one-to-four-year and five-or-more-year residence categories using migration histories from all SA2s other than the SA2 containing j. Denote these leave-one-SA2-out shares by $r^{1\text{–}4}_{a,t,-d(j)}$ and $r^{5+}_{a,t,-d(j)}$. The instrument is

$$Z^T_{jt} = \sum_a \left[s^{1\text{–}4}_{ja,t-5} r^{1\text{–}4}_{a,t,-d(j)} + s^{5+}_{ja,t-5} r^{5+}_{a,t,-d(j)} \right].$$

Thus, variation in the instrument comes from differences in lagged local age-by-tenure composition interacted with the residence patterns of otherwise similar age groups outside the focal SA2. Excluding the entire SA2 prevents the focal SA1's own migration outcomes from entering the retention rates applied to its exposure shares.

C.3 Demographic instrument for housing costs

Housing costs may respond to changes in local demand that also affect residential utility. To isolate variation in housing costs driven by predetermined demographics, we construct an instrument from each SA1's lagged age composition and the national age distribution of deaths. Let $s_{ja,t-5}$ be the share of residents in age group a in SA1 j five years before year t, and let q_a be the share of deaths nationally that occur in that age group. The instrument is

$$Z^p_{jt} = \sum_a s_{ja,t-5} q_a. \tag{74}$$

National weights are common across locations, so variation in Z^p_{jt} reflects differences in the age composition of local residents. Locations with greater exposure to age groups associated with deaths may experience different rates of housing turnover and, consequently, different pressure on rents. Lagging the local age shares separates this predicted demographic pressure from contemporaneous changes in housing costs. The identifying restriction is that, conditional on the controls and fixed effects in the residential-preference specification, this demographic exposure affects recovered flow utility through housing costs rather than through other changes in local amenities.

D Identification of the distribution of community water values

This section establishes sufficient conditions for nonparametric identification of the distribution of persistent community water values. We first take the conditional noise distribution as identified and prove the deconvolution result. We then give sufficient conditions under which repeated post-reform observations identify that noise distribution. The final part relates these conditions to the empirical estimator.

D.1 Signals and conditioning information

Let $t = 0$ denote 2006 and let $t = 1, 2, 3$ denote the three post-reform waves. Write $W_{jt} = \Delta W_{jt}$, with $W_{j0} = 0$. Once the common water effect and nuisance components have been removed, residualized utility levels satisfy

$$Y_{jt} = A_j + \eta_j W_{jt} + e_{jt}. \tag{75}$$

Here A_j is a time-invariant location component, $\omega_j = \overline{\omega} + \eta_j$, and $\mathbb{E}[\eta_j] = 0$. The mean normalization presumes a finite first moment. The pre–post differences are

$$r_{jt} \equiv Y_{jt} - Y_{j0} = \eta_j W_{jt} + u_{jt}, \qquad u_{jt} \equiv e_{jt} - e_{j0}. \tag{76}$$

The identification argument takes the mean effect and nuisance adjustments as identified population objects. Sampling uncertainty in their estimates is addressed by the bootstrap described separately.

Let X_j denote observed or identified conditioning information sufficient for the restrictions below. It may include water-district and the vector of first-stage residuals from

$$W_{jt} = \pi_t Z_j + \kappa_{d(j),t} + v_{jt}. \tag{77}$$

The instrument provides exposure variation after conditioning on selection controls. A linear first-stage residual is not automatically a valid *distributional* control: the required restrictions concern conditional distributions, rather than only conditional means. All variables are taken to have well-defined regular conditional distributions.

D.2 Deconvolution with an identified noise distribution

Fix a post-reform wave t. The following assumptions are sufficient; no normality, finite-mixture restriction, or density for η_j is required. For almost every x in the support of X_j, assume:

1. **Conditional independence.** (η_j, u_{jt}) is independent of W_{jt} conditional on $X_j = x$, and η_j and u_{jt} are independent conditional on $X_j = x$.

2. **Identified noise law.** The conditional distribution $F_{u_t|X=x}$ is identified from observable variation. Sufficient conditions based on repeated observations are provided below.

3. **Invertibility.** The conditional characteristic function $\varphi_{u_t|x}(s) \equiv \mathbb{E}[\exp(isu_{jt}) \mid X_j = x]$ is nonzero for every $s \in \mathbb{R}$.

4. **Nonzero exposure.** There is a nonzero conditional exposure w at which the conditional law of $r_{jt} \mid W_{jt} = w, X_j = x$ is identified. This can be a nonzero mass point. For continuous exposures, conditional laws are identified almost everywhere; a nonzero exposure in that identified set suffices. Continuity of the conditional characteristic function in exposure also permits evaluation at other support points.

Proposition D.1 (Nonparametric identification)**.** Under Assumptions 1–4, $F_{\eta|X=x}$ is identified for almost every x. Consequently, F_η and the distribution of $\omega_j = \overline{\omega} + \eta_j$ are identified when $\overline{\omega}$ is identified.

Proof. Fix an admissible x and an identified nonzero exposure w. For any $s \in \mathbb{R}$, Equation (76) and conditional independence imply

$$\varphi_{r_t|w,x}(s) = \mathbb{E}[\exp\{is(\eta_j w + u_{jt})\} \mid W_{jt} = w, X_j = x] \tag{78}$$

$$= \mathbb{E}[\exp(isw\eta_j) \mid X_j = x]\mathbb{E}[\exp(isu_{jt}) \mid X_j = x] \tag{79}$$

$$= \varphi_{\eta|x}(sw)\varphi_{u_t|x}(s). \tag{80}$$

For an arbitrary $a \in \mathbb{R}$, set $s = a/w$. Since the noise characteristic function is nonzero,

$$\varphi_{\eta|x}(a) = \frac{\varphi_{r_t|w,x}(a/w)}{\varphi_{u_t|x}(a/w)}. \tag{81}$$

Both terms on the right are identified. Thus the characteristic function of $\eta_j \mid X_j = x$ is identified on the entire real line. Uniqueness of characteristic functions identifies its probability distribution. Finally,

$$F_\eta(a) = \int F_{\eta|X}(a \mid x)\, dF_X(x), \tag{82}$$

where F_X is observed. Translating this distribution by the identified mean $\overline{\omega}$ identifies F_ω. □

The support requirement is modest once the noise law is identified: a single nonzero conditional exposure suffices for this argument. At zero exposure the signal contains only noise and provides no information about η_j. Additional exposure values provide further restrictions on the same conditional distribution. The nonvanishing condition is a sufficient invertibility condition, not a claim that every noise distribution satisfies it. Conditional independence is substantive and is stronger than the orthogonality conditions used to identify a mean effect by IV.

The result concerns the cross-community distribution. A finite number of noisy signals generally leaves uncertainty about any particular η_j. Predicting individual values requires a specification of how the available signals are combined; the distributional identification result alone does not specify a joint likelihood for those signals.

D.3 Identifying noise from repeated observations

Consider two post-reform waves s and t with stable exposure, $W_{js} = W_{jt} = W_j$. Equation (75) gives

$$r_{js} = Y_{js} - Y_{j0} = \eta_j W_j + (e_{js} - e_{j0}), \tag{83}$$

$$q_{jts} = Y_{jt} - Y_{js} = e_{jt} - e_{js}. \tag{84}$$

The post–post difference is observed and contains no persistent water component. Suppose, conditional on the same X_j used above,

$$e_{jt} - e_{js} \overset{d}{=} e_{js} - e_{j0}. \tag{85}$$

Then the observable conditional distribution of q_{jts} identifies the noise law $F_{u_s|X}$ needed for the pre–post signal r_{js}.

Proposition D.2 (Identification from stable post-reform exposure)**.** Suppose Equation (75) holds, post-reform exposure is stable between waves s and t, and Equation (85) holds. Under Assumptions 1, 3, and 4 for the pre–post signal r_{js}, $F_{\eta|X}$ and F_η are nonparametrically identified from the observable distribution of $(Y_{j0}, Y_{js}, Y_{jt}, W_j, X_j)$.

Proof. Stable exposure eliminates $\eta_j W_j$ from $Y_{jt} - Y_{js}$. Equation (85) equates its conditional distribution with that of $u_{js} = e_{js} - e_{j0}$, establishing Assumption 2. Proposition D.1 then applies. □

This argument does not require the pre–post and post–post noise realizations to be independent: they may share the shock e_{js}. It requires equality of the relevant conditional marginal noise laws. A strictly stationary level-shock process yields equal difference distributions over equal lags. Pooling five-year and ten-year differences requires an additional restriction; stationarity alone does not equate those distributions. If stable exposure holds only for a subset of communities, using their post–post differences for other communities additionally requires the same conditional noise law across those groups.

D.4 Relation to the empirical estimator

The estimator uses the actual wave-specific exposures. With changing exposure, the post–post difference is

$$Y_{jt} - Y_{js} = \eta_j(W_{jt} - W_{js}) + (e_{jt} - e_{js}). \tag{86}$$

Consequently, it is a pure noise observation only when the persistent water component vanishes. The empirical procedure pools the three post–post differences and estimates a Gaussian kernel density separately within seven quantile bins of the average first-stage residual. A Gaussian kernel is a smoothing choice; it does not impose a Gaussian noise distribution.

Using this pooled density as the pre–post noise law is an approximation when water exposure changes. It requires the water component in Equation (86) to be negligible and the relevant shock-difference distributions to be comparable across the pooled pairs and pre–post signals. Without these conditions, the pooled post–post density need not identify the noise distribution in Proposition D.1. Repeated observations alone do not justify treating it as that distribution.

Conditioning on seven bins of the average residual, after removing a linear wave-specific control function, also approximates the full conditioning information X_j. The theorem allows $F_{\eta|X}$ to vary with X; estimating a common F_η across bins imposes an additional restriction. The baseline

two-Gaussian family is an estimation restriction rather than a condition of the nonparametric identification result. Normal and grid-based nonparametric specifications provide comparisons under the same noise construction.

The numerical estimator first removes a preliminary IV water coefficient and allows the fitted mixing distribution to have an unrestricted mean. Centering that fitted distribution and absorbing its mean into $\overline{\omega}$ preserves total community coefficients and, holding the fitted noise densities fixed, all signal likelihoods. The post–post noise proxy is constructed before this recentering. Its potential water component therefore involves the preliminary coefficient deviations used at that step, rather than necessarily the centered η_j reported subsequently.

Inference and point identification are distinct. The marginal composite likelihood need not specify the joint dependence of a community's three signals, and location bootstrap weights are shared across its observations. Bootstrap intervals quantify sampling variation under the maintained resampling scheme; they do not establish the conditional independence or noise-stability restrictions used in the identification argument.

D.5 Choosing the number of Gaussian components

In Section 5.1.1, we model the idiosyncratic value of water η_j as being drawn from some known distribution F. In Section 6.1.3, we estimate F—assuming that it is a Gaussian mixture drawn with M components. In the limit, when M is large, this can flexibly approximate many underlying distributions.

For the purposes of estimation, we treat M as a hyper-parameter to be tuned. We face a trade-off between greater non-parametric flexibility and more over-fitting, with higher M. We choose M by K-fold cross validation over locations. We group locations within the same SA2 to not split SA1 locations between training and test data. For each fold k, we estimate $\hat{F}_{M,k}$ for M between 1 and 5. We then compute the held-out score on the test locations for each fold as,

$$CV_{M,k}\left(\hat{F}_{M,k}\right) = \sum_{j \in \text{test}(k)} \sum_{t \in \{2011,2016,2021\}} \log g_{jt}(\widetilde{r}_{jt}; \hat{F}_{M,k}), \tag{87}$$

where $g_{jt}(\widetilde{r}; F)$ is as in (34). We estimate this procedure with 50 folds across three independent trials. We compute the common components of the estimator which consist of everything outside of $\hat{F}_{M,k}$, using the full sample. In addition to this cross-validation score, we are able to compute the Bayesian Information Criterion (BIC) for robustness. In Table A2, we present the results of this analysis. We find that for both metrics, $M = 2$ is the best choice for the number of mixtures. However, the variance of η_j and the percentage of time that $\bar{\omega} + \eta_j < 0$ are relatively stable for all $M \geq 2$. We find that when $M = 1$, and F is a normal distribution, it is strictly worse than all specifications with $M \geq 2$, highlighting the value of using a more flexible distribution in this setting.

Table A2: Validation of Gaussian component choice

M	50 folds	BIC gap	Variance of η_j	$\mathbb{P}(\eta_j \leq 0)$
1	−5.67 (1.40)	158	0.098 [0.089, 0.117]	1.3% [0.4, 3.9]
2	**0**	**0**	0.168 [0.142, 0.220]	3.2% [2.0, 5.2]
3	−0.71 (0.28)	7	0.175 [0.153, 0.226]	3.5% [1.8, 5.5]
4	−0.21 (0.38)	23	0.174 [0.155, 0.227]	2.8% [1.7, 5.3]
5	−0.36 (0.38)	49	0.174 [0.155, 0.229]	2.7% [1.4, 5.4]

Notes: This table presents the K-fold cross-validation scores, BIC gap, and summary statistics for estimated distributions $\hat{F}$ under different assumptions of the number of Gaussian components, M. The variance and the probability that $\eta_j \leq 0$ report 95% confidence intervals from a Bayesian bootstrap, holding fixed estimates of $\hat{\zeta}$.

D.6 Testing for Roy-style selection on gains

If locations with high idiosyncratic values of η_j were to purchase more water, this would be consistent with a story of Roy-style selection on gains. It would also be consistent with a story in which η_j is correlated with V_j, in which the additional water purchases, beyond what would be predicted by soil characteristics, are directed towards locations with idiosyncratically large gains from water. We test for this with a linear version of Masten & Torgovitsky (2016), similar in spirit to Garen (1984) and Wooldridge (1997). We estimate the following regression, separately by bins of V_j,

$$\Delta\widehat{\zeta}_{j,2011,2006} = \beta W_j + \gamma_{d(j)}. \tag{88}$$

We present these parameter estimates in Figure A1. We find no systematic heterogeneity. In fact if anything, it seems as though the β estimates seem to be slightly smaller in the locations with large V_j.

E Construction of Regional Productive Water Values

This appendix describes how we adapt the farm-level production-function estimates of Rafey (2023) to our regional agricultural data. We do not re-estimate the underlying crop-specific production technologies. Instead, we take Rafey's preferred technological parameters as given and recover the regional objects required to evaluate the marginal product of irrigation water at observed regional allocations.

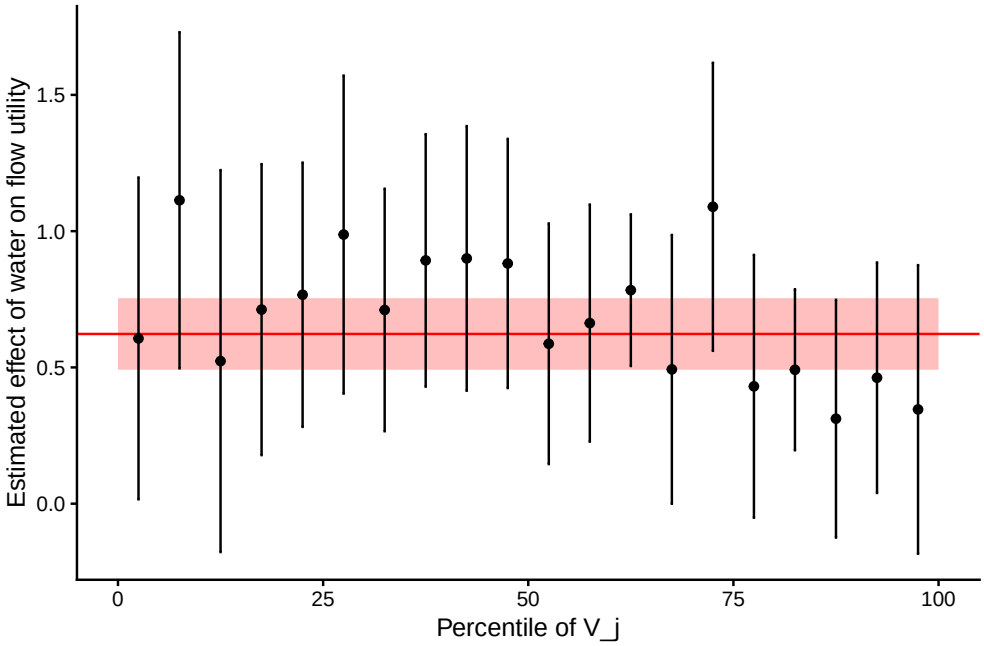
E.1 Production Technology

For crop c in region j and year t, production is

$$Y_{jct} = \exp\{\omega_{jct}\}\left[\alpha_c\left(W_{jct} + \theta_c R_{jct}\right)^{\frac{\sigma_c-1}{\sigma_c}} + (1-\alpha_c)K_{jct}^{\frac{\sigma_c-1}{\sigma_c}}\right]^{\frac{\sigma_c}{\sigma_c-1}\beta_{cW}} L_{jct}^{\beta_{jcL}}, \tag{89}$$

where W_{jct} is irrigation water, R_{jct} is rainfall, K_{jct} is cultivated land, L_{jct} is labor, and ω_{jct} is crop-region-year productivity. Since we do not separately observe output prices or expenditures on

Figure A1: Estimated Community Value of Water, by Quantile of V_j



Notes: This figure plots the estimated β as estimated in (88), separately estimated by ventile of V_j. The pooled estimate from the linear IV is overlaid in red.

intermediate inputs, ω_{jct} absorbs variation in these components in addition to physical productivity. We therefore interpret ω_{jct} as a crop-region-year revenue productivity shifter, consistent with our objective of recovering the marginal value of irrigation water.

The crop-specific parameters α_c, θ_c, σ_c, β_{cW}, and β_{cL} are taken from Rafey (2023). The parameter α_c determines the distributional share of water relative to land, θ_c converts rainfall into effective water, σ_c governs substitution between effective water and land, and β_{cW} and β_{cL} govern the output elasticities of the water-land aggregate and labor.

The regional data differ from the farm microdata used to estimate these parameters in one important respect. Aggregating across heterogeneous farms can cause the response of total regional labor to differ from the within-farm labor elasticity. We therefore allow the effective regional labor elasticity to deviate from the farm-level estimate:

$$\beta_{jcL} = \widehat{\beta}_{cL} \exp\{\delta_j\}, \tag{90}$$

where δ_j is a persistent SA2-specific adjustment common across crops.

E.2 Regional Labor Elasticities

Because production is Cobb–Douglas in labor, the first-order condition for labor implies

$$\beta_{jcL} = \frac{w_{jt} L_{jct}}{Y_{jct}}, \tag{91}$$

where w_{jt} is the agricultural wage.

We observe total agricultural labor L_{jt} at the regional level rather than its allocation across crops. Summing the crop-specific first-order conditions and substituting Equation (90) gives

$$\exp\{\delta_j\} = \frac{w_{jt} L_{jt}}{\sum_c \widehat{\beta}_{cL} Y_{jct}} \tag{92}$$

in the absence of measurement error or transitory deviations from the model.

Define the observed annual regional deviation

$$d_{jt} \equiv \log\left(\frac{w_{jt} L_{jt}}{\sum_c \widehat{\beta}_{cL} Y_{jct}}\right). \tag{93}$$

We allow this object to contain both a persistent regional component and a transitory deviation:

$$d_{jt} = \delta_j + \nu_{jt}, \qquad \nu_{jt} \sim \mathcal{N}(0, \sigma_\nu^2). \tag{94}$$

We discipline persistent regional departures from the farm-level estimates with the hierarchical prior

$$\delta_j \sim \mathcal{N}(0, \sigma_\delta^2). \tag{95}$$

Centering the distribution at zero centers the regional elasticity on Rafey's farm-level estimate while allowing the data to reveal persistent departures from it.

Let

$$\overline{d}_j = \frac{1}{T_j} \sum_t d_{jt} \tag{96}$$

denote the average observed deviation for region j. Conditional on the variance components, normality implies

$$\delta_j \mid \overline{d}_j \sim \mathcal{N}\left(\kappa_j \overline{d}_j,\ \left[\frac{1}{\sigma_\delta^2} + \frac{T_j}{\sigma_\nu^2}\right]^{-1}\right), \tag{97}$$

where

$$\kappa_j = \frac{\sigma_\delta^2}{\sigma_\delta^2 + \sigma_\nu^2 / T_j}. \tag{98}$$

The Empirical Bayes estimate is therefore the posterior mean

$$\widehat{\delta}_j = \widehat{\kappa}_j \overline{d}_j. \tag{99}$$

Regions for which the observed labor share is measured precisely receive little shrinkage, while noisier regional estimates are pulled more strongly toward the farm-level benchmark. We estimate σ_δ^2 and σ_ν^2 jointly by maximum marginal likelihood. For region j, stack the T_j annual signals into $\mathbf{d}_j = (d_{j1}, \ldots, d_{jT_j})'$. Integrating out the persistent regional component δ_j gives

$$\mathbf{d}_j \sim N\left(0,\ \Sigma_j\right), \qquad \Sigma_j = \sigma_\nu^2 I_{T_j} + \sigma_\delta^2 \mathbf{1}_{T_j} \mathbf{1}_{T_j}'. \tag{100}$$

We estimate the variance components by maximizing the resulting marginal likelihood across re-

gions. Equivalently, region j contributes

$$-\frac{1}{2}\left[T_j \log(2\pi) + (T_j - 1)\log\sigma_\nu^2 + \log\left(\sigma_\nu^2 + T_j\sigma_\delta^2\right) + \frac{\sum_t (d_{jt} - \bar{d}_j)^2}{\sigma_\nu^2} + \frac{T_j \bar{d}_j^2}{\sigma_\nu^2 + T_j\sigma_\delta^2}\right] \tag{101}$$

to the log likelihood. The within-region variation in d_{jt} identifies σ_ν^2, while persistent differences in regional means identify σ_δ^2.

The resulting regional crop-specific labor elasticities are

$$\widehat{\beta}_{jcL} = \widehat{\beta}_{cL} \exp\{\widehat{\delta}_j\}. \tag{102}$$

E.3 Crop-Region Productivity

Conditional on the regional labor elasticities, crop-region-year productivity is the residual required for Equation (89) to rationalize observed output. Define

$$A_{jct} = \widehat{\alpha}_c (W_{jct} + \widehat{\theta}_c R_{jct})^{\frac{\widehat{\sigma}_c - 1}{\widehat{\sigma}_c}} + (1 - \widehat{\alpha}_c) K_{jct}^{\frac{\widehat{\sigma}_c - 1}{\widehat{\sigma}_c}}. \tag{103}$$

Then productivity is recovered by inversion:

$$\widehat{\omega}_{jct} = \log Y_{jct} - \frac{\widehat{\sigma}_c}{\widehat{\sigma}_c - 1}\widehat{\beta}_{cW} \log A_{jct} - \widehat{\beta}_{jcL} \log L_{jct}. \tag{104}$$

Because agricultural employment is observed at the regional rather than crop level, we allocate regional labor across crops according to the model-implied labor shares,

$$L_{jct} = L_{jt} \frac{\widehat{\beta}_{cL} Y_{jct}}{\sum_{c'} \widehat{\beta}_{c'L} Y_{jc't}}. \tag{105}$$

E.4 Marginal Productive Value of Water

With the production parameters and productivity shifters in hand, we evaluate the marginal product of irrigation water at each observed crop-region-year allocation. Let

$$\rho_c \equiv \frac{\widehat{\sigma}_c - 1}{\widehat{\sigma}_c} \tag{106}$$

and

$$A_{jct} = \widehat{\alpha}_c (W_{jct} + \widehat{\theta}_c R_{jct})^{\rho_c} + (1 - \widehat{\alpha}_c) K_{jct}^{\rho_c}. \tag{107}$$

Differentiating Equation (89) with respect to irrigation gives

$$\widehat{V}_{jct}^P = \frac{\partial Y_{jct}}{\partial W_{jct}} = Y_{jct} \widehat{\beta}_{cW} \frac{\widehat{\alpha}_c (W_{jct} + \widehat{\theta}_c R_{jct})^{\rho_c - 1}}{A_{jct}}. \tag{108}$$

Because output is measured in value terms, this derivative gives the marginal contribution of an additional unit of irrigation to producer revenue and, by the envelope argument in Section 5, to optimized operating profits.

Finally, we aggregate crop-specific marginal values to a single region-year value using observed irrigation shares:

$$\widehat{V}^P_{jt} = \sum_c s^W_{jct} \widehat{V}^P_{jct}, \qquad s^W_{jct} = \frac{W_{jct}}{\sum_{c'} W_{jc't}}. \tag{109}$$

This corresponds to valuing a marginal proportional increase in regional irrigation while holding fixed the observed allocation of water across crops.

F Converting Community Water Values to Dollars

Our estimates of community water values are initially in utility units for a 10% change in applied irrigation. We convert these estimates into annual community willingness to pay in AUD per ML.

We first use the coefficient on log weekly housing costs, $\alpha_p < 0$, as the money metric for utility. If weekly housing costs in community j are p_j, one unit of flow utility corresponds to $52p_j/|\alpha_p|$ dollars per household per year. We aggregate across households using population N_j and average household size $\bar{h}$.

We then convert the normalized water measure into applied water. Let s_W denote the change in residual ET, in ML/ha, corresponding to a 10% change in applied irrigation. Our validation of residual ET against administrative data on regional water allocations implies that one additional ML of allocated water generates 0.65 ML of dry-season residual ET. Finally, because residual ET is measured per hectare, we use the mean SA1 area in our estimation sample, $\bar{A}$, to convert the intensity change into a total water quantity. We use a common rather than community-specific area so that geographic differences in administrative-unit size do not mechanically generate differences in per-ML community values.

Combining these conversions gives

$$V^C_j = \omega_j \left(\frac{52p_j}{|\alpha_p|}\right)\left(\frac{N_j}{\bar{h}}\right)\left(\frac{0.65}{s_W \bar{A}}\right), \tag{110}$$

where ω_j is community j's utility value of a 10% change in applied irrigation. Equation (110) yields community willingness to pay in AUD per ML of applied irrigation water.[24]

We apply the same linear transformation to the empirical-Bayes posterior distribution of ω_j. If C_j denotes the conversion factor multiplying ω_j in Equation (110), then

$$\widehat{V}^C_j = C_j \widehat{\omega}^{EB}_j, \qquad \mathrm{SD}(V^C_j \mid \text{data}) = |C_j| \,\mathrm{SD}(\omega_j \mid \text{data}). \tag{111}$$

G Take-It-or-Leave-It Compensation Offers

This appendix derives the take-it-or-leave-it (TIOLI) compensation mechanism used in Section 8. The problem is the procurement analogue of the optimal posted-price problem in Myerson (1981)

[24] Housing costs serve only as the numeraire for converting utility into dollars. Changes in housing costs induced by water reallocation are not separately counted as community welfare effects.

and Bulow & Roberts (1989): traders have a known surplus from completing the transaction but face a community whose reservation value is privately known.

Let $S > 0$ denote the surplus available from a potential trade. The selling community privately observes its value of retaining the water, V_s^C, while traders know its population distribution,

$$V_s^C \sim F_C, \tag{112}$$

with density f_C. Traders make a compensation offer p, which the community accepts whenever $V_s^C \leq p$. The probability that the trade is accepted is therefore $F_C(p)$. Conditional on acceptance, traders retain $S - p$, so their expected payoff is

$$\Pi(p; S) = (S - p)F_C(p). \tag{113}$$

The optimal offer solves

$$p^*(S) = \arg\max_{0 \leq p \leq S} (S - p)F_C(p). \tag{114}$$

For an interior solution, the first-order condition is

$$(S - p^*)f_C(p^*) = F_C(p^*), \tag{115}$$

or equivalently,

$$S - p^* = \frac{F_C(p^*)}{f_C(p^*)}. \tag{116}$$

The left-hand side is the surplus traders retain when the offer is accepted. A marginal increase in the offer reduces this surplus on every accepted trade, but increases the probability of acceptance by $f_C(p)$. The optimal offer balances these two effects. As in the monopoly-pricing interpretation of optimal auctions in Bulow & Roberts (1989), traders optimally shade the offer below the amount they could afford to pay because a higher offer increases the probability of trade at the cost of lower surplus conditional on trade. This informational friction can therefore prevent some trades for which $S \geq V_s^C$ and full compensation would have been feasible.

In our implementation, we approximate F_C by a Normal distribution using the estimated population mean and variance of selling-community values and solve Equation (114) numerically for each potential trade. We consider two definitions of available surplus. With private gains alone,

$$S = G_{sb}^P, \tag{117}$$

while allowing gains to the buying community to finance compensation gives

$$S = G_{sb}^P + \max\{V_b^C, 0\}. \tag{118}$$

The information available to traders about V_s^C is the same in both cases; buying-community gains increase only the surplus available to finance the offer. A trade is implemented when the realized selling-community value satisfies $V_s^C \leq p^*(S)$, in which case the community receives $p^*(S)$.

H Bootstrap inference

Our inference procedure propagates uncertainty in estimated residential utilities through the estimation of community water values and the counterfactual analysis. It combines resampling of household destination choices with location-level bootstrap weights. The first component perturbs the migration flows used to estimate utilities; the second changes the relative contribution of communities to the estimated distribution of water values. All subsequent transformations are recomputed using the corresponding estimates within each replication.

H.1 Residential utilities and demand

Let c index a household type, census wave, and origin–tenure state, and let n_{cj} denote the observed count choosing destination j. Write $N_c = \sum_j n_{cj}$ and $\widehat{p}_{cj} = n_{cj}/N_c$. For each replication b, we independently draw destination counts within each cell:

$$(n_{cj}^{(b)})_j \sim \text{Multinomial}\Big(\widetilde{N}_c, (\widehat{p}_{cj})_j\Big), \qquad \widetilde{N}_c = \text{round}(N_c). \tag{119}$$

Thus the bootstrap holds the size and choice set of each cell fixed while resampling its destination composition. With integer counts, this is equivalent to independently resampling observed destination choices within the cell. Because some input counts are fractional, we round cell totals for resampling while retaining the original count shares. Cells whose totals round to zero retain their original counts. The point estimator uses the original fractional counts.

For each resampled dataset, we re-estimate outside-option conditional choice probabilities, continuation terms, and the complete residential utility and moving-cost parameter vector. We preserve the original retained support, parameter definitions, initialization rule, and numerical stopping criteria. Zero-count utility groups remain in the estimator, with the finite values returned by the numerical optimization. This preserves the treatment of these groups in the point estimates; the resampling distribution consequently pertains to that numerical estimator, including its treatment of sparse choices.

We then re-estimate the demand IV regression using the corresponding utility estimates and the original regressors, instruments, fixed effects, and weights. Let $\widehat{\alpha}^{(b)}$ be the resulting coefficient on log rent. The utilities used in distribution estimation are net of price expenditure:

$$\widehat{\zeta}_{jt\tau}^{(b)} = \widehat{\omega}_{jt\tau}^{(b)} - \widehat{\alpha}^{(b)} \log(rent_{jt}). \tag{120}$$

We aggregate across tenure using the same shares as in the point estimation and maintain a common complete location panel. Each utility draw remains paired with its own estimated price coefficient throughout the analysis.

For the demand coefficients themselves, generated-utility uncertainty may already contribute to destination-clustered standard errors. We therefore use a variance-replacement calculation rather than simply adding the variance of the bootstrap coefficients. Let $\widehat{V}_{\text{cl}}$ be the original clustered covariance matrix. Centering the utility draws at their pointwise bootstrap means and passing these

perturbations through the fixed IV design yields coefficient perturbations with covariance $\widehat{V}_{\text{first}}$. Let $\widehat{V}_{\text{embedded}}$ be the average clustered covariance from those perturbation regressions, multiplied by $B/(B-1)$ to match the sample-covariance normalization. The adjustment is

$$\widehat{V}_{\text{adjusted}} = \widehat{V}_{\text{cl}} + \widehat{V}_{\text{first}} - \widehat{V}_{\text{embedded}}. \tag{121}$$

This calculation replaces the first-step contribution represented in the clustered covariance with its propagated covariance across utility estimates. Its interpretation requires orthogonality between centered first-step errors and latent demand disturbances and an adequate clustering specification for those disturbances. It is an approximate generated-outcome adjustment; positive covariance estimates alone do not establish its coverage properties. We do not introduce an additional independent draw of the price coefficient in the subsequent analysis.

H.2 The distribution of community values

For each residential-utility draw, we generate independent location weights

$$w_j^{(b,r)} = \frac{E_j^{(b,r)}}{J^{-1}\sum_{\ell=1}^{J} E_\ell^{(b,r)}}, \qquad E_j^{(b,r)} \overset{\text{iid}}{\sim} \text{Exp}(1). \tag{122}$$

These normalized exponential weights implement a Bayesian bootstrap over locations. A location receives the same weight in all census waves, retaining its complete history. Using these weights, we re-estimate the water first stages, control-function coefficients, district–year means, residual bins, noise densities, and the distribution of community water-value coefficients. The same weights are used for all distributional specifications. The main specification uses the two-Gaussian family selected by cross-validation; normal and nonparametric specifications provide comparisons. Model-family selection is held fixed in the bootstrap.

We use $B = 100$ residential-utility draws and $R = 10$ location-weight draws per utility draw, producing 1,000 paired replications. These replications contain 100 distinct residential-utility realizations, rather than 1,000 independent first-step estimates. Location-only replications, computed on the original utility panel, provide a separate comparison. Numerical convergence is recorded for each fit; intervals requiring all replications are not reported when any required fit is unavailable or nonconverged.

Write $a + \eta_j$ for a community's water-value coefficient in utility units and C_j for its fixed monetary-conversion factor. Within replication (b, r), its value in AUD/ML is

$$V_j^{(b,r)} = \frac{C_j}{|\widehat{\alpha}^{(b)}|}\big(\widehat{a}^{(b,r)} + \eta_j\big). \tag{123}$$

We recompute dollar values before calculating means, quantiles, and the fraction exceeding AUD 235/ML. The population distribution averages the fitted utility distribution after applying each community's conversion factor, with equal weight across communities. This construction assigns a common marginal utility distribution before conversion; it does not estimate additional dependence

between latent utility coefficients and conversion factors. The distribution of posterior means instead summarizes the community-specific conditional expectations. It is distinct from the population distribution because it omits remaining within-community posterior variation.

H.3 Posterior uncertainty and counterfactuals

Each replication supplies both a fitted population distribution and updated community posteriors, computed using the same signal likelihood and tempering rule as the point estimates. A posterior standard deviation measures remaining uncertainty about a community's latent value conditional on a fit. The bootstrap standard deviation of its posterior mean instead measures variation in that estimated conditional expectation across fits. We report these quantities separately.

For correlations with farmer marginal values and induced labor spending, we draw a value from each community's posterior, apply (123), and recompute the correlation. Community identities are retained, and a community's sampled value is reused across years and outcomes within a realization. Location weights enter the correlation calculation. Multiple posterior realizations within each fit provide a combined bootstrap–posterior uncertainty distribution. This is a model-based assessment of the latent relationship: the posteriors do not additionally condition on farmer values. Annual correlations use fixed matched samples, while pooled correlations stack their matched community–year observations.

Counterfactuals use fixed sets of within-year, non-overlapping, privately profitable trading opportunities, generated before the bootstrap and reused across mechanisms and replications. Within each fit, we sample a latent community value for every location and use the same realization in all mechanisms. Traders choose offers using the fitted population distribution consistent with their information: the seller-specific dollar distribution when its conversion factor is known, and a distribution averaged over selling opportunities when that factor is unknown. They do not observe the simulated seller value when choosing the offer. The simulated value determines acceptance. Full-information and uniform-compensation rules are evaluated on the same opportunities and community realizations.

We average counterfactual outcomes over posterior realizations within each fit, and then form bootstrap intervals across these fit-level averages. Consequently, these intervals concern posterior-averaged implementation, compensation, and welfare, rather than the outcome for a particular latent realization. Bootstrap pair weights equal the product of endpoint location weights, normalized within each fixed market. Markets receive equal weight within years, and pooled counterfactual summaries weight years equally. The matchings are not reoptimized or resampled under these weights.

H.4 Intervals and interpretation

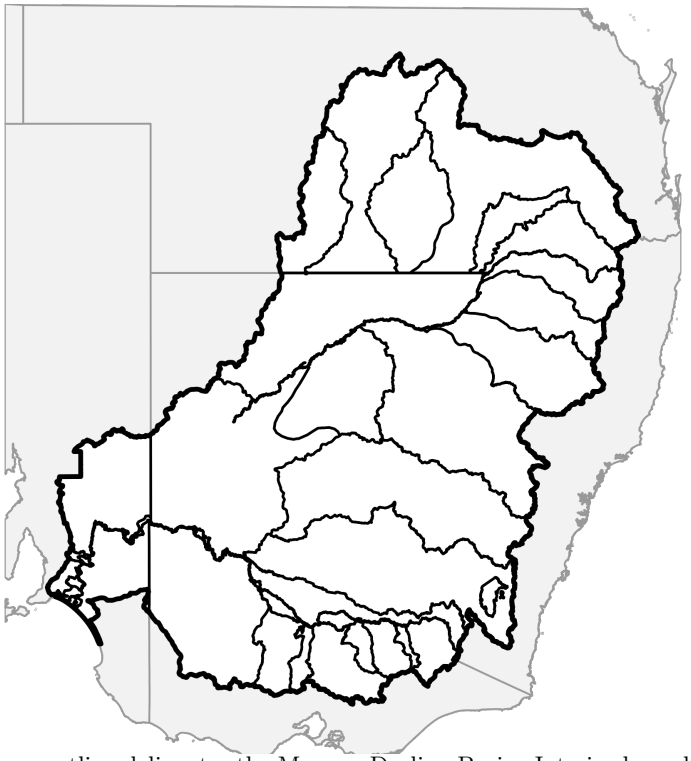
For each reported distributional statistic or posterior-averaged counterfactual, we compute the statistic separately in every paired replication. We report its sample standard deviation and the

2.5th and 97.5th percentiles of the replication distribution. Bands over uniform payments are pointwise, not simultaneous. When reporting outcomes at the trade-maximizing posted price, the main calculation selects the price from the baseline seller-veto curve and holds it fixed across replications. These intervals do not adjust for baseline price selection. Reselecting the maximizer within each replication provides a separate policy-stability exercise, rather than a validated confidence set for the optimal price.

The procedure conditions on water covariates, demand regressors and instruments, farmer-value estimates, and monetary-conversion factors. It treats destination choices as independent conditional on their cells and combines these perturbations independently with location weights. It does not model repeated-household dependence across waves or spatial dependence across locations. The reported intervals therefore reflect the maintained resampling and empirical-Bayes specifications; the two resampling components are not interpreted as separately identified additive variance contributions.

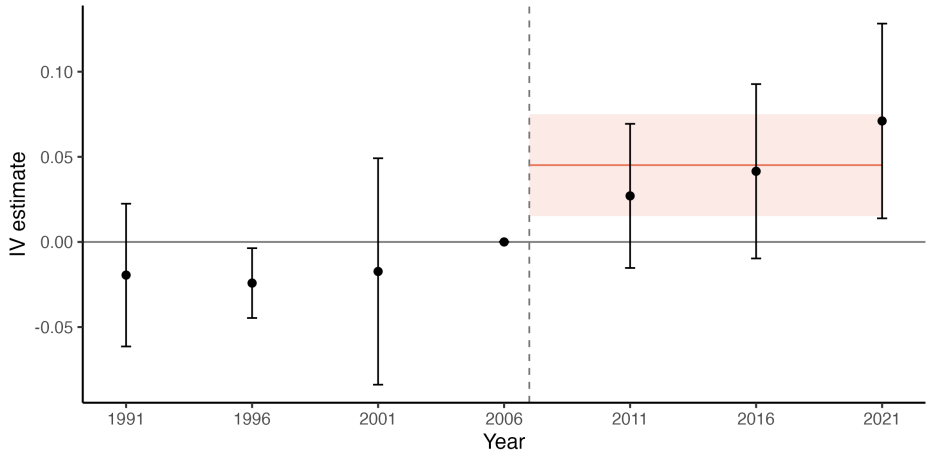
I Appendix Figures

Figure A2: Murray–Darling Basin Map and Water-Districts

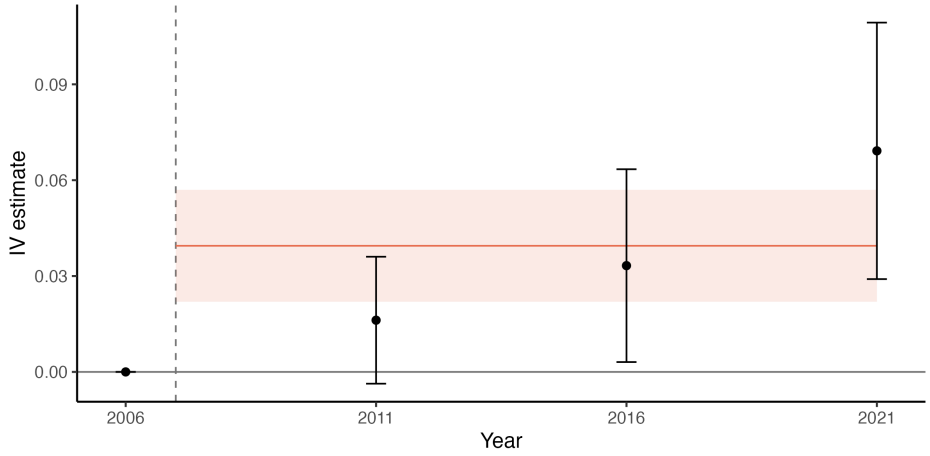


Notes: The heavy outline delineates the Murray–Darling Basin. Interior boundaries show the water-districts used to organize the administrative water data. State boundaries are shown in gray for geographic reference.

Figure A3: Effect of Water Gain on Population and Long-Term Residents



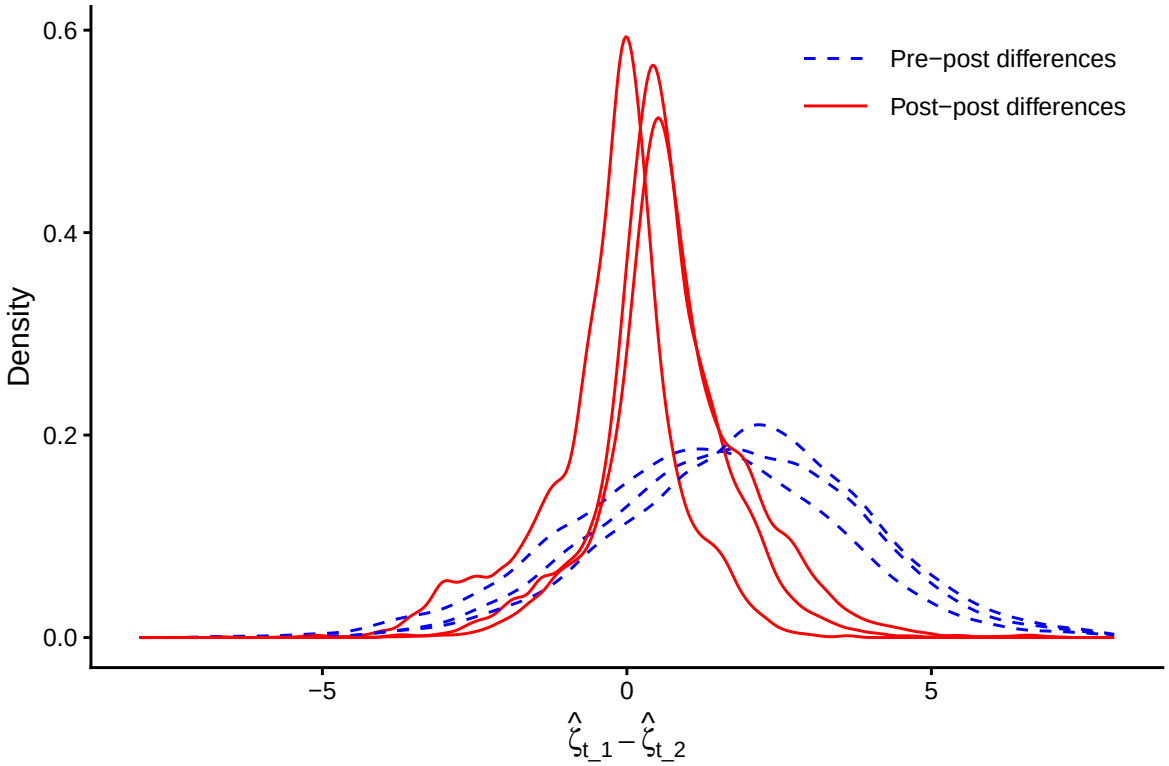
(a) Population



(b) Long-term residents

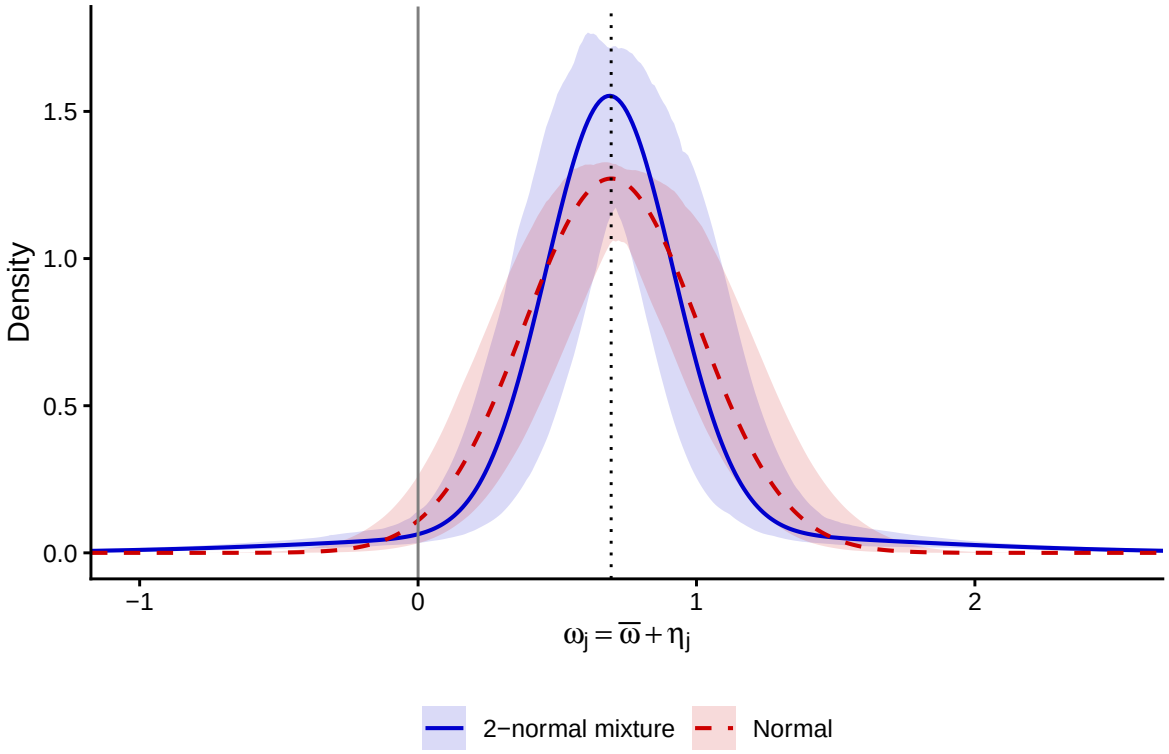
Notes: Figure depicts the effect of a persistent increase of 0.096 ET/hectare on SA1 population and the stock of long-term residents relative to 2006 for each Census year from 1991–2021 when data is available. We instrument for water change with soil characteristics. One unit of water change is equivalent to a 10% increase in irrigation intensity at the average SA1. Annual estimates are reported in black dots and the pooled post-reform effect is the horizontal red line. Standard errors are clustered at the SA2 level for annual results and at the SA1 level for the pooled effect.

Figure A4: Distribution of the Differences in Amenity Values Across Periods: $q_{j,t,s}$



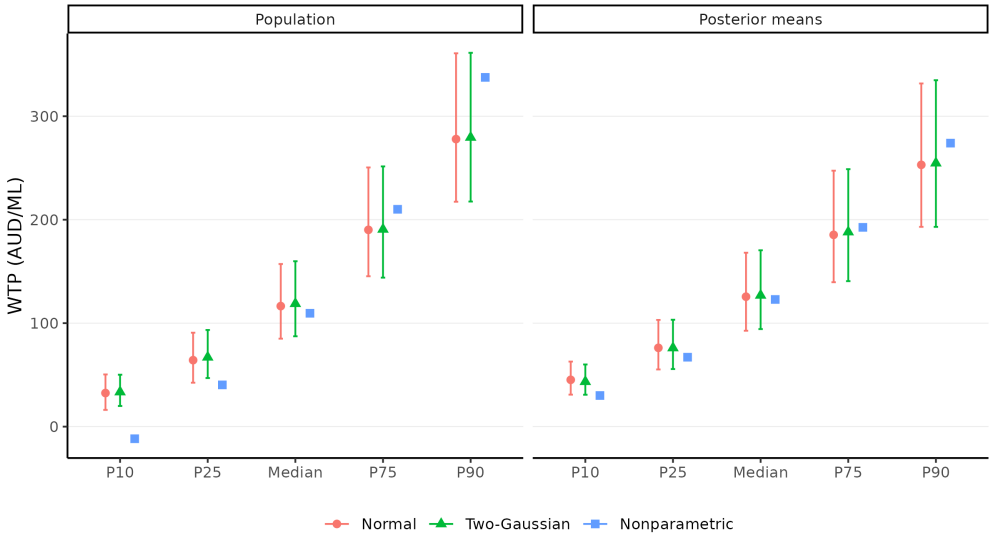
Notes: The figure plots the distribution of the estimated distribution of the difference in the amenity value of living in a location j between periods t and s. We display pre-post differences with blue dashed lines, and post-post differences with red solid lines.

Figure A5: Distribution of the Marginal Amenity Value of Water



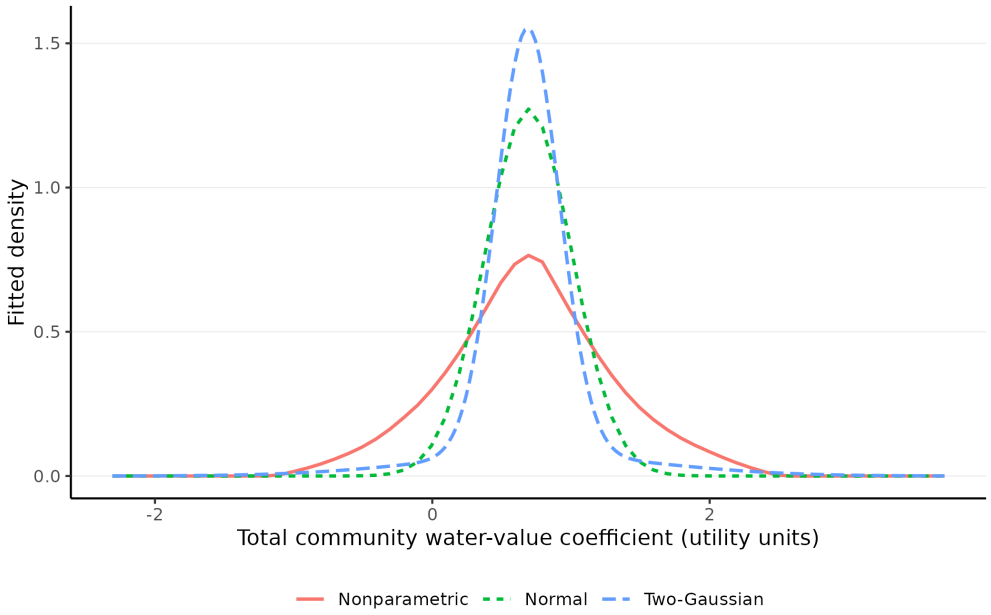
Notes: The figure plots the distribution of the estimated distribution of the location-specific idiosyncratic values for water, $\omega_j \sim \hat{F}$. The figure shows the two-normal mixture, as determined by the cross-validation procedure in Appendix Section D.5 in a solid blue line with 95% confidence interval bands. The normal distribution is the red dashed line.

Figure A6: Quantiles of Community Value Distribution across Specifications



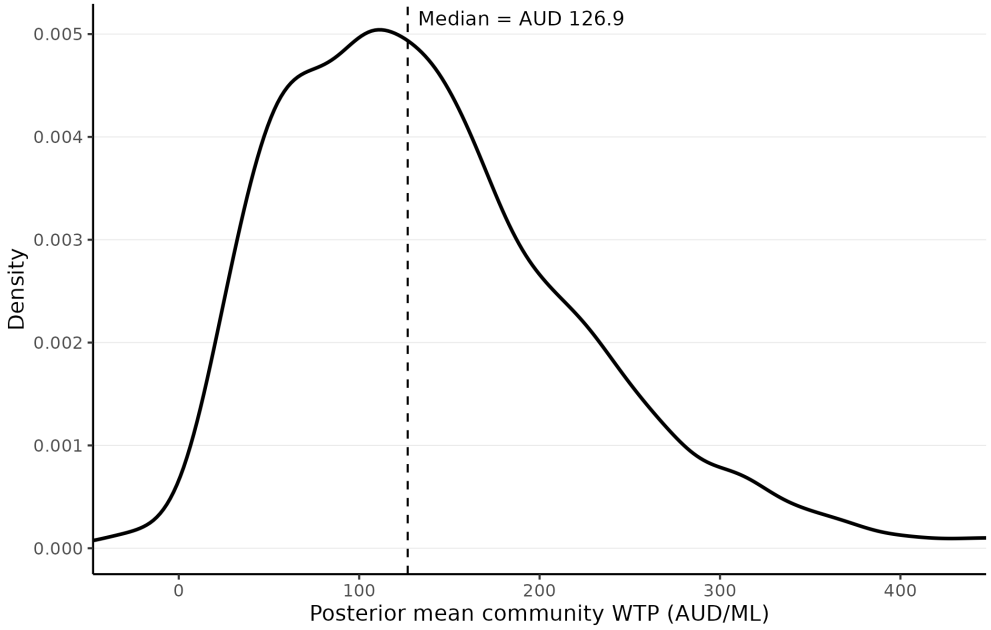
Notes: The figure plots quantiles of the estimated distribution under normality, two-Gaussian, and nonparametric fits. We report the quantiles for the estimated population distribution and the distribution of posterior means. Values are express in 2021 AUD per ML.

Figure A7: Distribution of Community Values across Specifications



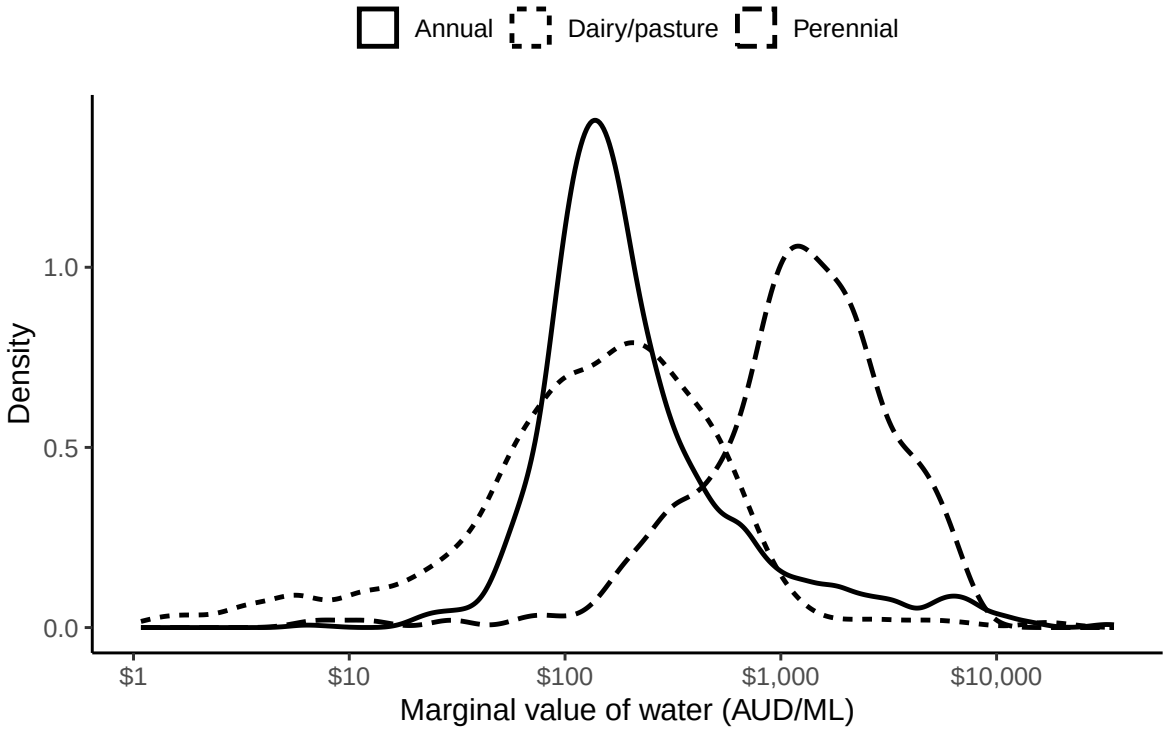
Notes: The figure plots the population distribution of mean community values for local irrigation across SA1s for different specifications of the distribution. Values are expressed in 2021 AUD per ML.

Figure A8: Distribution of Community Value Posterior Means



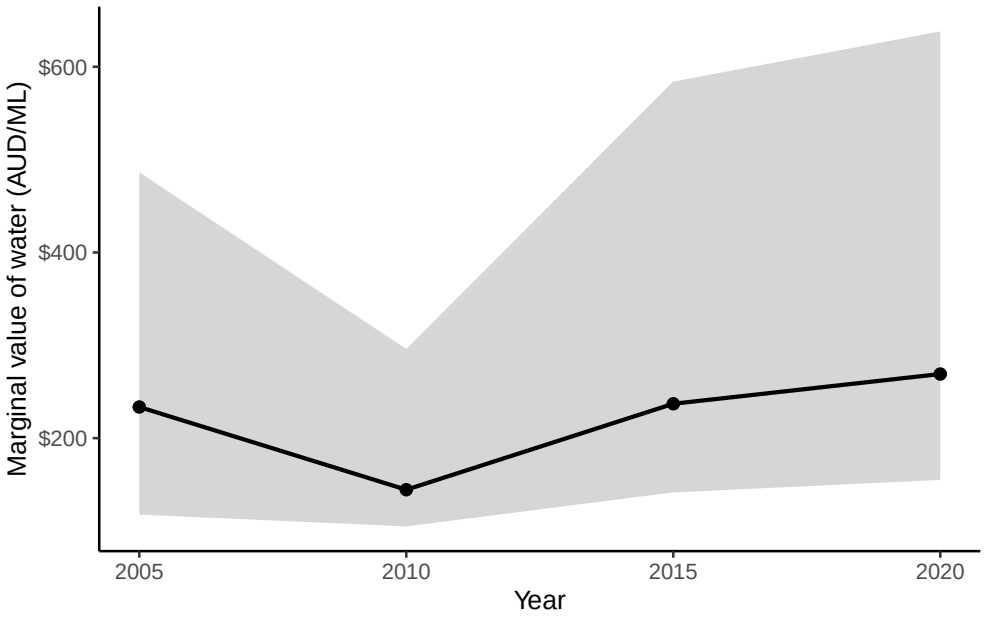
Notes: The figure plots the posterior mean community values for local irrigation across SA1s. Values are expressed in 2021 AUD per ML.

Figure A9: Distribution of the Marginal Productive Value of Water



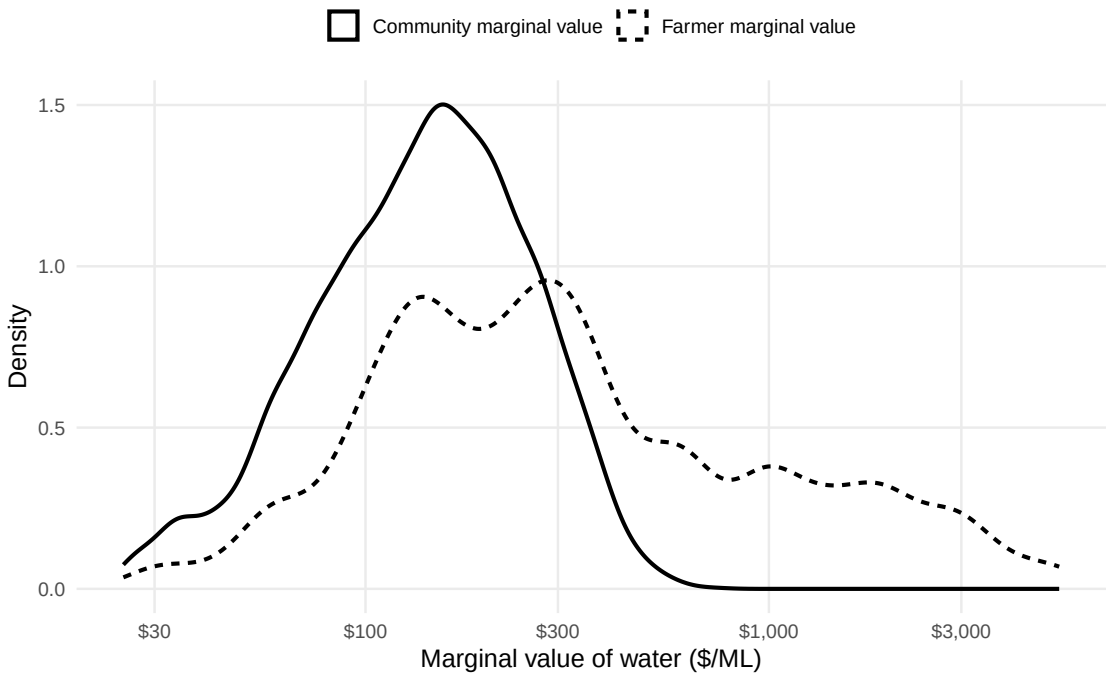
Notes: The figure plots the distribution of the estimated marginal value of applied irrigation water across crop–SA2–year observations. The horizontal axis is logarithmic. The estimation sample shown here excludes crop–SA2–year cells with fewer than 100 hectares of observed production. Marginal values are measured in AUD per megaliter.

Figure A10: Marginal Productive Value of Water over Time



Notes: The solid line reports the median water-weighted marginal value of irrigation water across SA2s in each year. The shaded region denotes the interquartile range. Crop-specific marginal values are aggregated within an SA2 using observed shares of applied irrigation water. Values are measured in AUD/ML.

Figure A11: Comparing private and community values for water



Notes: Each panel plots the distribution of the water-weighted marginal productive value of irrigation water across SA2s against the marginal community value of water across SA1s. The horizontal axis is logarithmic. Values are measured in AUD/ML.

Figure A12: Uniform Compensation without Community Consent



Notes: The figure traces outcomes under a uniform compensation requirement as the required payment to selling communities varies. The horizontal axis reports the share of privately gainful trade opportunities implemented, and the vertical axis reports the share of selling communities associated with implemented trades whose estimated willingness to pay exceeds the required payment. The solid curve allows compensation to be financed from private gains alone; the dashed curve additionally allows buying-community surplus to be mobilized. Circles denote the full-information community-veto benchmarks and triangles denote the private-information TIOLI benchmarks. Dollar labels report the uniform payment that produces the corresponding trade rate.

J Appendix Tables

Table A3: SA1-year Census Variable Summary

Variable	N	Mean	SD	P10	Median	P90
Long-tenure share	23,424	0.5	0.1	0.4	0.5	0.7
Household income	23,152	1,181.9	705.1	454.8	1,029.0	2,154.0
Population	23,782	349.8	147.9	192.0	344.0	523.0
Tenure stock	21,752	0.8	2.9	0.5	0.7	1.0
Weekly mortgage payment	23,094	1,323.3	586.1	699.3	1,300.0	2,059.0
Weekly rent	23,114	222.0	128.7	75.0	203.0	395.0

Notes: The table summarizes the SA1-year observations used to construct local outcomes and residential characteristics. N is the number of nonmissing SA1-year observations for each variable; SD is the standard deviation, and P10 and P90 are the 10th and 90th percentiles. Income, rent, and mortgage payments are reported in Australian dollars in their underlying Census reporting periods. Long-tenure share and tenure stock are proportions.

Table A4: First Stage: Agricultural Suitability and Water Reallocation

	2011 (1)	2016 (2)	2021 (3)	Pooled Post (4)
Root-zone water storage	-0.0066***	-0.0055***	-0.0052***	-0.0057***
	(0.0011)	(0.0009)	(0.0010)	(0.0004)
Water-region fixed effects	Yes	Yes	Yes	No
Water-region × year fixed effects	No	No	No	Yes
Observations	6,494	6,494	6,494	19,482
Excluded-instrument F-statistic	34.1	35.6	28.0	183.9

Notes: The dependent variable in each column is the change in local irrigation activity accumulated through the indicated year relative to the pre-reform period. Root-zone water storage is constructed from predetermined soil characteristics. Columns (1)–(3) include water-district fixed effects and report standard errors clustered by SA2 in parentheses. Column (4) pools the three post-reform years, includes water-district-by-year fixed effects, and clusters standard errors by SA1. The excluded-instrument F-statistic is the clustered Wald F-statistic for root-zone water storage.

Table A5: Effects of Water Reallocation on Local Outcomes

Outcome	Pooled estimate	95% confidence interval
Economic Activity		
Household Income	0.0306 (0.0056)	[0.0196, 0.0415]
Rent	0.0516 (0.0089)	[0.0341, 0.0690]
Mortgage	0.0513 (0.0088)	[0.0341, 0.0685]
Personal Income	0.0030 (0.0028)	[-0.0024, 0.0084]
Ag. Employment	0.0750 (0.0189)	[0.0380, 0.1120]
Total Businesses	0.0261 (0.0054)	[0.0156, 0.0366]
Population and residence		
Population	0.0451 (0.0152)	[0.0153, 0.0750]
Tenure Stock	0.0368 (0.0083)	[0.0204, 0.0532]
Political and social outcomes		
Labor Vote Share	-0.0215 (0.0103)	[-0.0417, -0.0013]
Suicide Rate	-0.1736 (0.0971)	[-0.3640, 0.0167]

Notes: The table reports pooled post-reform IV estimates of the effect of a persistent 0.096 ML/ha increase in dry-season residual evapotranspiration (ET). Water changes are instrumented using predetermined soil characteristics. This change in residual ET corresponds to approximately a 10% increase in irrigation intensity at the average SA1. Business outcomes, Labor vote share, and the suicide rate are observed at the SA2 level; all other outcomes are observed at the SA1 level. Standard errors, in parentheses, are clustered at the geographic level at which each outcome is observed.

Table A6: Robust Price Preference Estimate across Specifications

	Baseline	No irrigation	No long-term residents	Income + housing only
Log housing costs	−2.096	−2.240	−1.982	−2.059
	(0.528)	(0.591)	(0.334)	(0.330)
Irrigation	✓		✓	
Long-term resident stock	✓	✓		
Household income	✓	✓	✓	✓
Death Exposure Control	✓	✓	✓	✓
Destination FE	✓	✓	✓	✓
Water region × year FE	✓	✓	✓	✓
Tenure FE	✓	✓	✓	✓
Observations	59,284	59,284	59,284	59,284

Notes: The dependent variable is flow indirect utility recovered from the first-stage residential choice model. The specifications include tenure-state, destination, and water-district-by-year fixed effects and controls for the local death rate. Income, rent, tenure stock, and irrigation are instrumented as described in Section 6. Standard errors are clustered by destination and first-stage uncertainty is incorporated by bootstrapping households.

Table A7: Agricultural Production Given Parameters

Crop	α_c	σ_c	θ_c	β_{cW}	β_{cL}
Perennial	0.599	1.575	1.081	0.631	0.352
Annual	0.513	1.451	1.048	0.526	0.201
Dairy	0.385	5.000	0.148	0.782	0.147

Notes: The table reports crop-specific production parameters taken from Rafey (2023) and held fixed when applying the production function to our SA2 data. α_c is the share parameter for effective water in the water–land aggregate; σ_c is its elasticity of substitution; θ_c converts rainfall into effective water; and β_{cW} and β_{cL} are the output elasticities of the water–land aggregate and labor. For annual crops, we use the parameters estimated for irrigated production because our objective is to value an additional unit of applied irrigation at observed irrigated allocations.

Table A8: Distribution of Recovered Agricultural Productivity

crop	N	Mean	SD	p10	Median	p90
annual	8217	10.711	4.879	8.191	11.300	13.476
dairy	8298	7.009	3.747	4.640	7.468	9.534
perennial	5364	8.877	6.626	6.690	9.435	11.669

Notes: The table summarizes the crop–SA2–year revenue productivity shifter $\widehat{\omega}_{jct}$ recovered by inverting the production function at observed output and inputs. The shifter absorbs differences in output prices and unobserved intermediate inputs as well as physical productivity; its levels should therefore not be interpreted as comparable measures of physical productivity across crops. N counts crop–SA2–year observations.

Table A9: Agricultural Production Estimated Parameters

	Mean	SD	p25	Median	p75
$\exp\{\hat{\delta}_j\}$	0.727	1.119	0.356	0.652	1.273

Notes: The table summarizes $\exp\{\widehat{\delta}_j\}$ across SA2s. This factor multiplies the crop-specific farm-level labor elasticity taken from Rafey (2023) to obtain the effective regional labor elasticity, $\widehat{\beta}_{jcL} = \widehat{\beta}_{cL} \exp\{\widehat{\delta}_j\}$. A value of one leaves the farm-level elasticity unchanged; values below or above one imply a lower or higher regional elasticity, respectively. SD is the standard deviation, and P25 and P75 are the 25th and 75th percentiles.

Table A10: Distribution of the Marginal Productive Value of Water

Crop	N	Mean	p25	Median	p75
Annual	2016	597	111	167	323
Dairy/pasture	2349	344	64	153	319
Perennial	621	1735	705	1258	2285
All crops, water-weighted	705	565	122	215	509

Notes: Entries report summary statistics for the estimated marginal value of applied irrigation water, in AUD/ML. Crop-specific statistics are calculated across crop–SA2–year observations with at least 100 hectares of production. The final row aggregates crop-specific marginal values within each SA2–year using each crop's share of applied water as weights.

Table A11: Administrative Water Components and Residual Evapotranspiration

Residual ET	Annual	Dry	Wet
Model:	(1)	(2)	(3)
Variables			
Direct allocation	0.7268*	0.6518*	-0.0387
	(0.4073)	(0.3380)	(0.1354)
Environmental buyback	-0.5730	-0.6834	0.0996
	(3.743)	(2.262)	(1.016)
Net agricultural transfer	0.7336	0.5954	0.0970
	(0.9230)	(0.6637)	(0.3374)
Fixed-effects			
Region	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Observations	440	440	440
R^2	0.31990	0.78178	0.93051

Notes: The dependent variables are annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. Administrative water availability is decomposed into direct allocations, environmental buybacks, and net agricultural transfers. All specifications include water-district and year fixed effects. Standard errors are clustered at the water-district level.

Table A12: Validation of Residual ET with Applied Irrigation Water

Dependent Variable:	Applied Water		
Model:	(1)	(2)	(3)
Variables			
Residual ET	1.19*	1.68**	-1.10
	(0.693)	(0.843)	(1.83)
Fixed-effects			
SA2	Yes	Yes	Yes
Water district×Year	Yes	Yes	Yes
Fit statistics			
Observations	1,170	1,170	1,170
R^2	0.720	0.721	0.719

Notes: The dependent variable is administrative applied irrigation water at the SA2-year level. Columns report specifications using annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. All specifications include SA2 fixed effects and water-district-by-year fixed effects. Standard errors are clustered at the SA2 level.